vibsynth 2.2 — Theory manual

Mission synthesis, kurtosis control, MMVSD and the compression limit, explained step by step

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Abstract

This manual explains every concept used by the `vibsynth` package, from the definition of a power spectral density to the Moment-Matched Variable Spectral Density (MMVSD) method introduced in version 2.1. Each quantity is defined before it is used, and every derivation is written out in full, so that the text can be followed by reading alone. Part I builds the common ground (random signals, the single-degree-of-freedom oscillator, fatigue damage, the Fatigue Damage Spectrum). Part II describes the mission synthesis and kurtosis control methods of version 2.0. Part III derives the MMVSD method, which prescribes, in addition to the Fatigue Damage Spectrum, the kurtosis of the response of every oscillator, and its treatment of sine-on-random blocks. Part IV, new in version 2.2, asks how far a record can be shortened at all: the damage Lorenz curve and the compression limit, the `edit` method that keeps the damaging stretches of the record and is exact for every S–N exponent up to that limit, the kurtosis a compressed test should carry, and the two-frequency description of non-stationarity behind the response kurtosis spectrum. The appendices contain the longer integrations (Lalanne's operator, the ramp-invariant oscillator filter) and the formulas quoted from the literature.

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1 Purpose, scope and notation

1.1 What the package does

A component mounted on a vehicle, a machine or a package is shaken during its life. The acceleration at its base is measured (a *record*), or prescribed by a *specification*, or described by a *mission*: a list of situations (road types, operating conditions) with the time spent in each. An accelerated vibration test reproduces, on a shaker and in a shorter time T, the fatigue damage that the life of duration T_{L} would cause. `vibsynth` computes the acceleration time history $x(t)$ of such a test (the *test profile*).

The fatigue damage is described by the Fatigue Damage Spectrum (FDS, Section 4.3): the damage that the vibration causes in a family of single-degree-of-freedom oscillators with different natural frequencies. Two signals with the same FDS damage every lightly damped component in the same way, whatever its natural frequency. Every synthesis method of the package (Parts II and III) therefore starts from the target FDS and builds a signal with that FDS; the methods differ in what they prescribe besides the FDS: the kurtosis of the signal, its RMS value, its power spectral density, tonal components, or, in MMVSD, the kurtosis of the response of every oscillator. The `edit` method of Part IV takes the opposite route: it does not synthesise but keeps the damaging stretches of the record itself, which makes it right for every S–N exponent, up to a limit that the record sets.

1.2 How to read this manual

The manual is written so that it can be followed without paper and pencil. Each symbol is defined at the place where it first appears, and each derivation shows all intermediate steps. Results that are used later are collected in boxes like this one:

Example of a result box

A boxed equation is a result that later sections refer to.

Three results are taken from the literature without derivation, because they are an empirical fit or classical theorems: Dirlik's distribution of rainflow ranges, the central limit theorem and the statistical properties of Welch's PSD estimate (Appendix C). They are stated precisely and their role is explained. Everything else is derived here.

1.3 Notation

t	time [s]
$f,\ \omega = 2\pi f$	frequency [Hz] and angular frequency [rad/s]
$x(t)$	base acceleration: the record or the test profile [m/s^2]
$f_s,\ \Delta t = 1/f_s$	sampling frequency and sampling interval
$N,\ T = N\Delta t$	number of samples and duration of a signal
T_L	life represented by a target (the duration of use that the test must reproduce)
$\mathbb{E}[\cdot]$	expectation (average over all realisations of a random process)
$\langle\cdot\rangle$	time average over the duration of a signal, $\langle g\rangle = \frac{1}{T}\int_0^T g(t)\,\mathrm{d}t$
$\sigma^2,\ \sigma$	variance and standard deviation (RMS value of a zero-mean signal)
κ	kurtosis (Pearson definition, 3 for a Gaussian variable); $\kappa_z(f_\mathrm{n})$ is called `k_z` in the program
$G(f)$	one-sided power spectral density (PSD) [(m/s^2)^2/Hz]
$f_\mathrm{n},\ \omega_\mathrm{n} = 2\pi f_\mathrm{n}$	natural frequency of an oscillator
$Q,\ \zeta = 1/(2Q)$	quality factor and damping ratio
$z(t)$	relative displacement of an oscillator (base minus mass) [m]
$H(f)$	frequency response from base acceleration to relative displacement
$b,\ C,\ K$	Basquin exponent, Basquin constant, stress per unit relative displacement
$D(f_\mathrm{n})$	Fatigue Damage Spectrum (FDS)
$\mathrm{MRS}(f_\mathrm{n})$	Maximum Response Spectrum
$\Gamma(\cdot)$	Euler's Gamma function, $\Gamma(s) = \int_0^\infty u^{s-1}e^{-u}\,\mathrm{d}u$

The natural frequencies at which the FDS is evaluated form a grid $f_{\mathrm{n},1} < f_{\mathrm{n},2} < \cdots < f_{\mathrm{n},n}$ (the *nodes*). By default the grid has 12 points per octave: $f_{\mathrm{n},i} = f_\mathrm{min}\,2^{(i-1)/12}$, the last node being moved exactly to f_max. A linear grid with a constant step is also available. An index i (or j) always runs over the nodes, so that a quantity defined at every node, such as $D_i = D(f_{\mathrm{n},i})$, is a vector of length n.

1.4 Organisation

Part I (Sections 2–4) introduces random signals, the oscillator and fatigue damage. Part II (Sections 5–6) explains how a profile with a prescribed FDS is designed, synthesised and corrected, and the methods of version 2.0. Part III (Sections 7–9) derives MMVSD and its treatment of sine-on-random blocks. Section 10 relates the theory to the settings of the program, and Section 11 defines the diagnostics and checks of the interface. Part IV (Sections 12–15, added in version 2.2 after the practical sections so that the numbering of versions 2.0 and 2.1 is unchanged) derives the compression limit of a record and the `edit` method, the kurtosis a compressed test should carry, and the non-stationarity matrix behind the response kurtosis spectrum.

Part I — Signals, oscillators and damage

2 Random signals

2.1 Statistics of a signal

Let $x_1, \dots, x_N$ be the samples of a signal. Its *mean* and its *central moments* of order m are

$$\bar{x} = \frac{1}{N}\sum_{k=1}^{N} x_k, \qquad \mu_m = \frac{1}{N}\sum_{k=1}^{N} (x_k - \bar{x})^m. \tag{2.1}$$

The *variance* is $\sigma^2 = \mu_2$, the standard deviation is $\sigma = \sqrt{\mu_2}$, and for a zero-mean signal σ is the root mean square (RMS) value. The *skewness* μ_3/σ^3 is zero for a signal whose values are distributed symmetrically about the mean. The *kurtosis* (Pearson's definition, used everywhere in this manual and in the program) is

$$\kappa = \frac{\mu_4}{\mu_2^2}. \tag{2.2}$$

It measures the weight of the tails of the distribution of the values: rare large values raise μ_4 much more than μ_2. The *crest factor* is $\max_k |x_k - \bar{x}|/\sigma$.

The same definitions apply to a random variable X when the averages over the samples are replaced by expectations: $\mu_m = \mathbb{E}[(X - \mathbb{E}X)^m]$.

2.2 The Gaussian distribution and its kurtosis

A zero-mean Gaussian random variable X with variance σ^2 has the probability density

$$p(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-x^2/(2\sigma^2)}. \tag{2.3}$$

Its even moments follow from one integration by parts. For an integer $m \geq 0$, write $x^{m+2}e^{-x^2/(2\sigma^2)} = x^{m+1}\cdot\left(x\,e^{-x^2/(2\sigma^2)}\right)$ and note that $x\,e^{-x^2/(2\sigma^2)}$ is the derivative of $-\sigma^2 e^{-x^2/(2\sigma^2)}$. Then

$$\begin{aligned}
\mathbb{E}[X^{m+2}] &= \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} x^{m+1}\, x\, e^{-x^2/(2\sigma^2)}\,\mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}\sigma}\Big[-\sigma^2 x^{m+1} e^{-x^2/(2\sigma^2)}\Big]_{-\infty}^{\infty} + \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} (m+1)x^m\,\sigma^2 e^{-x^2/(2\sigma^2)}\,\mathrm{d}x \\
&= 0 + (m+1)\,\sigma^2\,\mathbb{E}[X^m].
\end{aligned} \tag{2.4}$$

The bracket vanishes because the exponential decays faster than any power grows. Starting from $\mathbb{E}[X^0] = 1$: $\mathbb{E}[X^2] = 1\cdot\sigma^2\cdot 1 = \sigma^2$ and $\mathbb{E}[X^4] = 3\sigma^2\mathbb{E}[X^2] = 3\sigma^4$. Therefore

Kurtosis of a Gaussian variable

$$\mathbb{E}[X^4] = 3\sigma^4, \qquad \kappa = \frac{3\sigma^4}{(\sigma^2)^2} = 3. \tag{2.5}$$

A signal with $\kappa > 3$ is called *leptokurtic*: it has more large values than a Gaussian signal with the same RMS. Measured road and transport vibration is typically leptokurtic, with κ between 4 and 8, because its level changes with the road surface and contains shocks.

2.3 Power spectral density

A record of duration T sampled at f_s can be written exactly as a finite Fourier series. With $\Delta f = 1/T$ and the frequencies $f_m = m\,\Delta f$, $m = 0, 1, \ldots, N/2$,

$$x(t_k) = \bar{x} + \sum_{m=1}^{N/2} A_m \cos(2\pi f_m t_k + \varphi_m), \qquad t_k = k\Delta t, \tag{2.6}$$

where $A_m \geq 0$ is the amplitude and φ_m the phase of the line f_m. A cosine of amplitude A_m has mean square value $A_m^2/2$ over a whole number of periods (because $\cos^2\theta = \tfrac{1}{2} + \tfrac{1}{2}\cos 2\theta$ and the second term averages to zero), and two lines of different frequencies are orthogonal over the record (the average of their product is zero). The variance of the record is therefore the sum of the powers of its lines,

$$\sigma^2 = \sum_m \frac{A_m^2}{2}. \tag{2.7}$$

The *one-sided power spectral density* (PSD) distributes this power over frequency: the power of the line f_m is spread over the band Δf that the line represents,

$$G(f_m) = \frac{A_m^2}{2\,\Delta f}, \qquad \sigma^2 = \sum_m G(f_m)\,\Delta f \;\approx\; \int_0^{f_s/2} G(f)\,\mathrm{d}f. \tag{2.8}$$

The unit of G is $(\mathrm{m/s^2})^2/\mathrm{Hz}$. For a random process the PSD is the expectation of this quantity when $T \to \infty$; for a single record it must be *estimated*, because the amplitudes of individual lines scatter strongly from one record to another. The program uses Welch's estimate: the record is cut into segments of length $1/r$ (r is the requested resolution in Hz) overlapping by 50 %, each segment is multiplied by a Hamming window (a raised-cosine taper that reduces the leakage of power between lines), the squared moduli of their Fourier transforms are scaled as in (2.8) and averaged. Averaging many segments reduces the scatter; this is the only property of the estimate that the manual uses.

2.4 Gaussian signals with a prescribed PSD

Equation (2.6) also tells how to *build* a signal with a prescribed PSD $G(f)$: choose the amplitudes

$$A_m = \sqrt{2\,G(f_m)\,\Delta f} \tag{2.9}$$

and phases φ_m drawn independently and uniformly in $[0, 2\pi)$. This is a *random-phase multisine*. Its variance is $\sum_m A_m^2/2 = \sum_m G(f_m)\Delta f$, exactly the area under the prescribed PSD, for every choice of the phases. The value $x(t)$ at a given time is a sum of many independent terms $A_m\cos(2\pi f_m t + \varphi_m)$, each with zero mean and bounded; by the central limit theorem the distribution of such a sum approaches a Gaussian distribution when many lines carry comparable power. In practice a test of 60 s has a line spacing of 1/60 Hz, and hundreds of lines fall in any band of interest, so the multisine is Gaussian for all purposes.

A second construction is exactly Gaussian for any number of lines. Write each line as $A_m\cos(2\pi f_m t + \varphi_m) = U_m\cos(2\pi f_m t) - V_m\sin(2\pi f_m t)$ with $U_m = A_m\cos\varphi_m$ and $V_m = A_m\sin\varphi_m$ (this is the addition formula of the cosine). If U_m and V_m are drawn as independent zero-mean Gaussian variables with variance $G(f_m)\Delta f$, then $x(t)$ is, at every t, a sum of Gaussian variables with fixed coefficients, hence Gaussian, with variance $\sum_m G(f_m)\Delta f\,(\cos^2 + \sin^2) = \sum_m G(f_m)\Delta f$. The amplitude $A_m = \sqrt{U_m^2 + V_m^2}$ is then random instead of fixed: it is the length of a vector with two independent Gaussian components, whose distribution (the Rayleigh distribution) is derived in Section 4.4, with $\mathbb{E}[A_m^2/2] = G(f_m)\Delta f$.

The pair (U_m, V_m), written as the complex number $U_m + \mathrm{j}V_m$, is a *complex normal coefficient*; the program draws such coefficients with unit mean square modulus and scales them to the power of each line.

Keeping the phases (or the coefficients) of a realisation fixed while changing the amplitudes A_m is called *frozen phases*: the signal then changes smoothly with its PSD, which is what the closed-loop corrections of Section 5.4 rely on.

2.5 Modulated Gaussian signals

A leptokurtic signal can be built by multiplying a stationary Gaussian *carrier* $g(t)$ (zero mean, variance σ_g^2) by a positive *envelope* $e(t)$ that varies slowly compared with the carrier:

$$x(t) = e(t)\, g(t). \tag{2.10}$$

Divide the duration T into windows short enough for e to be practically constant in each (window w has duration T_w and envelope value e_w) but long enough to contain many periods of the carrier, so that the time average of g^2 and g^4 over a window equals its expectation. Using (2.5) for the carrier,

$$\left\langle x^2 \right\rangle = \sum_w \frac{T_w}{T}\, e_w^2\, \sigma_g^2 = \sigma_g^2 \left\langle e^2 \right\rangle, \tag{2.11}$$

$$\left\langle x^4 \right\rangle = \sum_w \frac{T_w}{T}\, e_w^4\, 3\sigma_g^4 = 3\sigma_g^4 \left\langle e^4 \right\rangle, \tag{2.12}$$

and the kurtosis (2.2) of the product is

Kurtosis of a modulated Gaussian signal

$$\kappa = \frac{3\sigma_g^4 \left\langle e^4 \right\rangle}{\left(\sigma_g^2 \left\langle e^2 \right\rangle\right)^2} = 3\,\frac{\left\langle e^4 \right\rangle}{\left\langle e^2 \right\rangle^2} = 3\,\frac{\left\langle s^2 \right\rangle}{\left\langle s \right\rangle^2}, \qquad s(t) = e(t)^2 \sigma_g^2. \tag{2.13}$$

The quantity $s(t)$ is the *local variance* of the signal. The last form of (2.13) states a general rule used throughout Part III: if, given a slowly varying local variance $s(t)$, the signal is Gaussian, its kurtosis is three times the ratio of the mean square of s to the square of its mean. By the Cauchy–Schwarz inequality, $\langle s \rangle^2 \le \left\langle s^2 \right\rangle \langle 1 \rangle$, so $\kappa \ge 3$, with equality only for a constant s. Modulation can raise the kurtosis but never lower it below the Gaussian value.

3 The single-degree-of-freedom oscillator

3.1 Equation of motion

A mass m is connected to a moving base by a spring of stiffness k_s and a viscous damper of coefficient c. The base moves with displacement $y(t)$, so its acceleration is $\ddot{y}(t) = x(t)$, the signal of interest. Let $u(t)$ be the absolute displacement of the mass and

$$z(t) = y(t) - u(t) \tag{3.1}$$

the *relative displacement* (base minus mass). The spring and the damper act on the mass with the force $k_s(y-u) + c(\dot{y}-\dot{u}) = k_s z + c\dot{z}$, so Newton's law reads $m\ddot{u} = k_s z + c\dot{z}$. Since $u = y - z$, $\ddot{u} = x - \ddot{z}$, and

$$m(x - \ddot{z}) = k_s z + c\dot{z} \quad \Longrightarrow \quad \ddot{z} + \frac{c}{m}\dot{z} + \frac{k_s}{m} z = x(t). \tag{3.2}$$

With the *natural angular frequency* $\omega_\mathrm{n} = \sqrt{k_s/m} = 2\pi f_\mathrm{n}$ and the *damping ratio* $\zeta = c/(2m\omega_\mathrm{n})$,

Oscillator under base acceleration

$$\ddot{z} + 2\zeta\omega_\mathrm{n}\dot{z} + \omega_\mathrm{n}^2 z = x(t), \qquad Q = \frac{1}{2\zeta}. \tag{3.3}$$

The quality factor Q is the usual way to give the damping in vibration testing; $Q = 10$ ($\zeta = 5\,\%$) is a common default. The stress in a component modelled by the oscillator is proportional to the deformation of its spring, $S = Kz$ with a constant K. The *pseudo-acceleration* $\omega_\mathrm{n}^2 z$ is often plotted instead of z: at low frequency it equals the base acceleration.

3.2 Frequency response

Let the base acceleration be the complex exponential $x(t) = e^{\mathrm{j}\omega t}$. The steady-state response has the same time dependence, $z(t) = H(\omega)e^{\mathrm{j}\omega t}$, with $\dot{z} = \mathrm{j}\omega z$ and $\ddot{z} = -\omega^2 z$. Substituting in (3.3):

$$\big(-\omega^2 + 2\mathrm{j}\zeta\omega_\mathrm{n}\omega + \omega_\mathrm{n}^2\big)H(\omega) = 1 \quad\Longrightarrow\quad H(\omega) = \frac{1}{\omega_\mathrm{n}^2 - \omega^2 + 2\mathrm{j}\zeta\omega_\mathrm{n}\omega}, \tag{3.4}$$

and, since $|a + \mathrm{j}b|^2 = a^2 + b^2$,

Frequency response

$$|H(\omega)|^2 = \frac{1}{(\omega_\mathrm{n}^2 - \omega^2)^2 + (2\zeta\omega_\mathrm{n}\omega)^2}. \tag{3.5}$$

At $\omega = 0$, $|H| = 1/\omega_\mathrm{n}^2$ (static deflection). At $\omega = \omega_\mathrm{n}$ the first term of the denominator vanishes and $|H(\omega_\mathrm{n})| = 1/(2\zeta\omega_\mathrm{n}^2) = Q/\omega_\mathrm{n}^2$: the resonance amplifies the static response by Q.

Half-power bandwidth. Near the resonance put $\omega = \omega_\mathrm{n}(1 + \varepsilon)$ with $|\varepsilon| \ll 1$. Then $\omega_\mathrm{n}^2 - \omega^2 = -\omega_\mathrm{n}^2(2\varepsilon + \varepsilon^2) \approx -2\varepsilon\omega_\mathrm{n}^2$ and $2\zeta\omega_\mathrm{n}\omega \approx 2\zeta\omega_\mathrm{n}^2$, so $|H|^2 \approx 1/\big(4\omega_\mathrm{n}^4(\varepsilon^2 + \zeta^2)\big)$. This is half of the peak value $1/(4\zeta^2\omega_\mathrm{n}^4)$ when $\varepsilon^2 + \zeta^2 = 2\zeta^2$, i.e. $\varepsilon = \pm\zeta$. The band between the two half-power points has the width $2\zeta\omega_\mathrm{n}$, or in hertz

$$\Delta f = 2\zeta f_\mathrm{n} = \frac{f_\mathrm{n}}{Q}. \tag{3.6}$$

An oscillator with $Q = 10$ responds mainly to a band of $\pm 5\,\%$ around f_n, i.e. about 0.14 octave wide. With 12 nodes per octave, adjacent oscillators ($2^{1/12} - 1 \approx 6\,\%$ apart) share most of their band. This overlap appears repeatedly in the design problems of Parts II and III.

3.3 Response to a random input

Let x be a stationary random signal with PSD $G(f)$, represented by the lines of Section 2.3. Each line $A_m\cos(2\pi f_m t + \varphi_m)$ produces the response line $|H(2\pi f_m)|A_m\cos(2\pi f_m t + \varphi_m + \arg H)$, so the response has the line powers $|H|^2 A_m^2/2$, and by (2.8) its PSD is

$$G_z(f) = |H(2\pi f)|^2\, G(f). \tag{3.7}$$

Its variance and, more generally, its *spectral moments* are

$$m_k = \int_0^\infty f^k\, G_z(f)\,\mathrm{d}f, \qquad k = 0, 1, 2, 4, \qquad \sigma_z^2 = m_0. \tag{3.8}$$

Differentiation multiplies a line by $\mathrm{j}\omega$, so the velocity $\dot{z}$ has the PSD $(2\pi f)^2 G_z$ and the variance $\sigma_{\dot{z}}^2 = (2\pi)^2 m_2$; likewise $\sigma_{\ddot{z}}^2 = (2\pi)^4 m_4$. These moments describe how often the response crosses zero and how many peaks it has (Section 4.4).

White noise. If $G(f) = G_0$ is constant, $\sigma_z^2 = G_0 \int_0^\infty |H|^2 \mathrm{d}f$. Appendix A shows that $\int_0^\infty |H|^2 \mathrm{d}f = 1/(8\zeta\omega_\mathrm{n}^3)$, hence

$$\sigma_z^2 = \frac{G_0}{8\zeta\omega_\mathrm{n}^3} = \frac{G_0\, Q}{4\omega_\mathrm{n}^3}, \qquad \sigma_{\omega_\mathrm{n}^2 z} = \sqrt{\omega_\mathrm{n}^4 \sigma_z^2} = \sqrt{\tfrac{\pi}{2}\, f_\mathrm{n}\, Q\, G_0}, \tag{3.9}$$

the second form being Miles' equation for the RMS pseudo-acceleration (use $\omega_\mathrm{n} = 2\pi f_\mathrm{n}$: $\omega_\mathrm{n}^4 G_0 Q/(4\omega_\mathrm{n}^3) = 2\pi f_\mathrm{n} G_0 Q/4 = \frac{\pi}{2} f_\mathrm{n} Q G_0$).

3.4 The response variance is linear in the PSD: Lalanne's operator

In the program a PSD is described by its values G_j at the nodes $f_{\mathrm{n},j}$ and is linear between consecutive nodes. Because (3.8) is linear in G, the response variance of the oscillator i is a linear combination of the nodal values,

Lalanne's operator

$$\sigma_{z,i}^2 = \sum_{j=1}^{n} W_{ij}\, G_j, \qquad W_{ij} = \int_0^\infty |H_i(f)|^2\, \phi_j(f)\, \mathrm{d}f, \tag{3.10}$$

where ϕ_j is the "hat" function equal to 1 at $f_{\mathrm{n},j}$, 0 at the other nodes and linear in between, and H_i is the response of the oscillator $f_{\mathrm{n},i}$. Lalanne integrated (3.10) in closed form; Appendix A gives the complete derivation. The matrix W is computed once per damage model. Two details of the program matter:

- the PSD of a synthesised profile is not cut at the first and last node but held constant over a margin of $3/Q$ relative bandwidth beyond them (an oscillator at the edge of the grid would otherwise see only half of its band); the methods of version 2.0 leave the effect of this margin to the closed loop, while the operators of MMVSD include it (Section 7.3);
- when the pre-filter of Section 3.7 is active, $|H_i|^2$ is multiplied by its power response.

3.5 How fast the response follows a change of the input

Part III needs to know how the response variance follows an input whose level changes with time. The mechanical energy of the oscillator per unit mass is $E = \frac{1}{2}\dot{z}^2 + \frac{1}{2}\omega_\mathrm{n}^2 z^2$. Its derivative, using (3.3) to replace $\ddot{z}$, is

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{z}\ddot{z} + \omega_\mathrm{n}^2 z\dot{z} = \dot{z}\,(\ddot{z} + \omega_\mathrm{n}^2 z) = \dot{z}\,(x - 2\zeta\omega_\mathrm{n}\dot{z}) = \dot{z}\,x - 2\zeta\omega_\mathrm{n}\dot{z}^2. \tag{3.11}$$

Take expectations: $\mathrm{d}\mathbb{E}[E]/\mathrm{d}t = \mathbb{E}[\dot{z}x] - 2\zeta\omega_\mathrm{n}\mathbb{E}[\dot{z}^2]$. Two facts close this equation.

Equipartition. In a lightly damped oscillator the response is a sinusoid at ω_n with a slowly varying amplitude A; over a cycle $\dot{z}^2$ and $\omega_\mathrm{n}^2 z^2$ have the same mean $\omega_\mathrm{n}^2 A^2/2$, so $\mathbb{E}[\dot{z}^2] = \omega_\mathrm{n}^2 \mathbb{E}[z^2] = \mathbb{E}[E]$.

Input power. The term $\mathbb{E}[\dot{z}(t)x(t)]$ is the power that the base injects. The velocity is a weighted sum of past inputs, $\dot{z}(t) = \int_0^\infty h_v(u) x(t-u)\,\mathrm{d}u$ (h_v is the velocity response to a unit impulse of x), so $\mathbb{E}[\dot{z}(t)x(t)] = \int_0^\infty h_v(u)\,\mathbb{E}[x(t-u)x(t)]\,\mathrm{d}u$. For a broadband input the correlation $\mathbb{E}[x(t-u)x(t)]$ vanishes after a time much shorter than one period of the oscillator: the injected power depends only on the current intensity of the input near f_n, not on the energy already stored. Its value follows from the stationary case, where $\mathrm{d}\mathbb{E}[E]/\mathrm{d}t = 0$: then $\mathbb{E}[\dot{z}x] = 2\zeta\omega_\mathrm{n}\mathbb{E}[E] = 2\zeta\omega_\mathrm{n}^3\sigma_z^2$, and with (3.9), $\mathbb{E}[\dot{z}x] = 2\zeta\omega_\mathrm{n}^3\, G_0/(8\zeta\omega_\mathrm{n}^3) = G_0/4$.

Let now the input intensity near f_n change slowly, $G_0 \to G(t)$, and write $s(t) = \mathbb{E}[z^2] = \mathbb{E}[E]/\omega_\mathrm{n}^2$ for the *local variance* of the response. Dividing the energy balance by ω_n^2:

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{G(t)}{4\omega_\mathrm{n}^2} - 2\zeta\omega_\mathrm{n}\, s = \frac{1}{\tau}\big(s_\infty(t) - s(t)\big), \qquad s_\infty(t) = \frac{G(t)}{8\zeta\omega_\mathrm{n}^3}, \tag{3.12}$$

where $s_\infty(t)$ is the stationary variance for the current input (3.9) and

Envelope law of the response

$$\tau = \frac{1}{2\zeta\omega_\mathrm{n}} = \frac{Q}{2\pi f_\mathrm{n}}, \qquad s(t) = \int_0^\infty \frac{1}{\tau} e^{-u/\tau}\, s_\infty(t-u)\,\mathrm{d}u. \tag{3.13}$$

The second form is the solution of the first-order equation (differentiate it to check). The response variance is the stationary variance low-pass filtered with the time constant τ. For a non-white input the same law is applied with $s_\infty(t) = \int |H|^2 G(f,t)\mathrm{d}f$ computed from the local spectrum $G(f,t)$.

3.6 Consequence: fast modulations do not reach the response

Suppose the stationary variance $s_\infty(t)$ fluctuates about its mean with variance V and an exponential correlation $\mathbb{E}[\delta s_\infty(t)\delta s_\infty(t+u)] = Ve^{-|u|/T_c}$, T_c being the correlation time (the duration of a burst). In the frequency domain the fluctuation has the two-sided spectrum $2VT_c/(1+\omega^2T_c^2)$ (its integral over $\omega/2\pi$ is V), and the filter of (3.13) has $|H_\tau(\omega)|^2 = 1/(1+\omega^2\tau^2)$. The variance of the filtered fluctuation is

$$\mathrm{Var}[s] = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \frac{2VT_c}{(1+\omega^2T_c^2)(1+\omega^2\tau^2)} = \frac{2VT_c}{2\pi}\cdot\frac{\pi}{T_c+\tau} = V\,\frac{T_c}{T_c+\tau}, \tag{3.14}$$

where the integral $\int_{-\infty}^{\infty} \mathrm{d}\omega/\big((1+\omega^2a^2)(1+\omega^2b^2)\big) = \pi/(a+b)$ follows from the partial fractions $\frac{1}{(1+\omega^2a^2)(1+\omega^2b^2)} = \frac{1}{a^2-b^2}\big(\frac{a^2}{1+\omega^2a^2} - \frac{b^2}{1+\omega^2b^2}\big)$ and $\int \mathrm{d}\omega/(1+\omega^2a^2) = \pi/a$: $\frac{1}{a^2-b^2}(\pi a - \pi b) = \pi/(a+b)$. The mean of s is not changed by the filter (its gain at $\omega = 0$ is 1). Inserting in the last form of (2.13), $\kappa = 3\langle s^2\rangle / \langle s\rangle^2 = 3(1 + \mathrm{Var}[s]/\langle s\rangle^2)$, the kurtosis of the response is

Low-pass effect of the oscillator on the kurtosis

$$\kappa_z - 3 = \big(\kappa_\infty - 3\big)\,\frac{T_c}{T_c+\tau}, \tag{3.15}$$

where $\kappa_\infty - 3 = 3V/\langle s\rangle^2$ is the excess kurtosis that the response would have if it followed the input instantaneously. Bursts much shorter than τ leave a nearly Gaussian response (this is Papoulis' rule); bursts much longer than τ are fully transmitted. Since τ is inversely proportional to f_n, the same input can produce Gaussian responses at low natural frequencies and leptokurtic ones at high natural frequencies. The kurtosis of the response, $\kappa_z(f_\mathrm{n})$, is therefore a function of the natural frequency, and it is this function, not the kurtosis of x, that governs the damage (Section 4).

3.7 The oscillator as the program computes it

The program computes z from the samples x_k with a second-order recursion,

$$z_k = b_1x_k + b_2x_{k-1} + b_3x_{k-2} - a_1z_{k-1} - a_2z_{k-2}, \tag{3.16}$$

whose coefficients are those of the *ramp-invariant* discretisation: they give the exact response of (3.3) at the sampling instants when x varies linearly between samples. Appendix B derives

them. The frequency response of the recursion is obtained by inserting $x_k = e^{\mathrm{j}\omega k\Delta t}$, $z_k = H_d e^{\mathrm{j}\omega k\Delta t}$:

$$H_d(f) = \frac{b_1 + b_2 e^{-\mathrm{j}\theta} + b_3 e^{-2\mathrm{j}\theta}}{1 + a_1 e^{-\mathrm{j}\theta} + a_2 e^{-2\mathrm{j}\theta}}, \qquad \theta = 2\pi f \Delta t. \tag{3.17}$$

It equals H when $f_\mathrm{n} \ll f_s$ but not when f_n approaches $f_s/2$. The ratio of the response variances to a flat input, $\int |H_d|^2 / \int |H|^2$, is for $Q = 10$:

f_s/f_n	25	8.8	6.25	4.4	3.3
variance ratio	0.99	0.92	0.85	0.72	0.56

The FDS of a record and of a test are both computed with (3.16), so the comparison is consistent; but a design based on the analogue (3.5) overestimates the response at high natural frequencies when f_s is only a few times f_max. The methods of version 2.0 design with H and let the closed loop (Section 5.4) correct the difference; MMVSD designs with H_d directly (Section 7.3).

Pre-filter. When $f_s \le 10 f_\mathrm{max}$ the program first passes x through a fixed third-order digital filter, forward and then backward in time (zero phase), as in the FDS computation of the dissertation. Applied twice, the filter multiplies the PSD by $|H_\mathrm{pre}(f)|^4$. The same filter is applied to records and tests, and $|H_\mathrm{pre}|^4$ multiplies $|H|^2$ in every operator of the program, so that the designs stay consistent with the measurement.

4 Fatigue damage and the Fatigue Damage Spectrum

4.1 Basquin's law and Miner's rule

A material loaded by cycles of constant stress amplitude S fails after $N(S)$ cycles. In the range of interest the S–N curve is a straight line in logarithmic scales (Basquin's law):

$$N(S)\, S^b = C, \tag{4.1}$$

with the exponent b (typically 3 to 12; 8 is a common default for structural details) and the constant C. A cycle of amplitude S consumes the fraction $1/N(S) = S^b/C$ of the life. Miner's rule adds these fractions: a history containing n_c cycles of amplitude S_c has the damage

$$D = \sum_c \frac{n_c}{N(S_c)} = \frac{1}{C} \sum_c n_c S_c^b, \tag{4.2}$$

and failure is expected when D reaches 1. With $S = Kz$ (Section 3), the damage of an oscillator is a sum over the cycles of its relative displacement:

$$D = \frac{K^b}{C} \sum_c n_c\, A_c^b, \qquad A_c = \text{amplitude (half range) of cycle } c. \tag{4.3}$$

Only the ratio of two damages matters in test tailoring, so K and C are set to 1 by default.

4.2 Rainflow counting

A random response does not consist of clean cycles; *rainflow counting* decomposes it into cycles in the way that is consistent with the stress–strain hysteresis of the material. The program proceeds in two steps.

1. *Turning points.* Only the local maxima and minima of the sampled response matter; the samples in between and plateaus are removed, leaving an alternating sequence $s_1, s_2, \dots$.
2. *Four-point rule.* Take four consecutive turning points $s_l, s_{l+1}, s_{l+2}, s_{l+3}$ and the three ranges between them. If the inner range $|s_{l+2} - s_{l+1}|$ is not larger than either outer range, the inner pair forms a closed cycle (a small loop inside a larger one): it is counted with amplitude $|s_{l+2} - s_{l+1}|/2$ and the two points are removed. The rule is applied until no more cycles close. The remaining sequence (the *residue*) is counted as half cycles, one per range, with weight $n_c = \frac{1}{2}$.

The program applies the four-point rule to all disjoint candidate pairs of the sequence at once, which is fast in NumPy, and finishes long monotone decays with a sequential stack. Its damage sums are equal to those of the sequential three-point algorithm of the ASTM E1049 standard (checked to 10^{-10} in the tests).

4.3 Fatigue Damage Spectrum and Maximum Response Spectrum

FDS and MRS

For a base acceleration x of duration T, the Fatigue Damage Spectrum is the damage (4.3) of the rainflow cycles of the response z_{f_n} of each oscillator of the grid,

$$D(f_\mathrm{n}) = r\,\frac{K^b}{C} \sum_{c \,\in\, \mathrm{rainflow}(z_{f_\mathrm{n}})} n_c A_c^b, \tag{4.4}$$

and the Maximum Response Spectrum is the largest pseudo-acceleration,

$$\mathrm{MRS}(f_\mathrm{n}) = \omega_\mathrm{n}^2 \max_t |z_{f_\mathrm{n}}(t)|. \tag{4.5}$$

The factor r is the number of times the signal is played: a record that represents a life T_L longer than its duration T has $r = T_\mathrm{L}/T$ (rounded to an integer when $T_\mathrm{L} \geq T$, i.e. the record is considered repeated), and a test profile that is shorter than the test duration is played r times (Section 10). Damage adds over repetitions; the maximum does not, so the MRS has no factor r. The MRS is a check of the peak stresses: a test whose MRS exceeds that of the life by much may fail the component by overload rather than by fatigue.

4.4 Rice's formulas for a Gaussian response

For a stationary Gaussian response the expected damage can be computed from the spectral moments.

Zero up-crossing rate. The response crosses zero upwards in the interval $(t, t + \mathrm{d}t)$ if $z(t) \in (-\dot{z}\,\mathrm{d}t, 0)$ with $\dot{z} > 0$. The probability of this event is $\int_0^\infty p(0,\dot{z})\,\dot{z}\,\mathrm{d}t\,\mathrm{d}\dot{z}$, where $p(z,\dot{z})$ is the joint density of z and $\dot{z}$. For a stationary process z and $\dot{z}$ are uncorrelated, because $\mathbb{E}[z\dot{z}] = \frac{1}{2}\,\mathrm{d}\mathbb{E}[z^2]/\mathrm{d}t = 0$, and being jointly Gaussian they are independent: $p(0,\dot{z}) = \frac{1}{\sqrt{2\pi}\sigma_z} \cdot \frac{1}{\sqrt{2\pi}\sigma_{\dot{z}}} e^{-\dot{z}^2/(2\sigma_{\dot{z}}^2)}$. Dividing the probability by $\mathrm{d}t$ and using $\int_0^\infty \dot{z}\, e^{-\dot{z}^2/(2\sigma_{\dot{z}}^2)}\mathrm{d}\dot{z} = \sigma_{\dot{z}}^2$,

$$\nu_0^+ = \frac{\sigma_{\dot{z}}^2}{2\pi\sigma_z\sigma_{\dot{z}}} = \frac{\sigma_{\dot{z}}}{2\pi\sigma_z} = \sqrt{\frac{m_2}{m_0}}, \tag{4.6}$$

using $\sigma_{\dot{z}} = 2\pi\sqrt{m_2}$ and $\sigma_z = \sqrt{m_0}$ (Section 3.3). The same reasoning applied to $\dot{z}$ (a maximum of z is a down-crossing of zero by $\dot{z}$) gives the rate of peaks $\nu_p = \sqrt{m_4/m_2}$. The ratio $\gamma =$

$\nu_0^+/\nu_p = m_2/\sqrt{m_0 m_4}$ is the *irregularity factor*: $\gamma = 1$ means one peak per zero crossing (narrow band).

Amplitudes of a narrow-band response. The response of a lightly damped oscillator is narrow band: a sinusoid at f_n whose amplitude changes slowly, $z = A\cos(\omega_\mathrm{n} t + \psi)$. Then $\dot z/\omega_\mathrm{n} \approx -A\sin(\omega_\mathrm{n} t + \psi)$ and the amplitude is $A = \sqrt{z^2 + (\dot z/\omega_\mathrm{n})^2}$. At a given time z and $\dot z/\omega_\mathrm{n}$ are independent zero-mean Gaussian variables with the same variance $\sigma^2 = \sigma_z^2$ (equipartition, Section 3.5). In polar coordinates (A, ψ) their joint density $\frac{1}{2\pi\sigma^2} e^{-A^2/(2\sigma^2)}$ times the Jacobian A, integrated over ψ, gives the *Rayleigh* density of the amplitude:

$$p(A) = \frac{A}{\sigma^2}\, e^{-A^2/(2\sigma^2)}, \qquad A \geq 0. \tag{4.7}$$

Its moment of order b follows with the substitution $u = A^2/(2\sigma^2)$, $A = \sqrt{2\sigma^2 u}$, $\mathrm{d}u = A\,\mathrm{d}A/\sigma^2$:

$$\mathbb{E}[A^b] = \int_0^\infty A^b\, \frac{A}{\sigma^2} e^{-A^2/(2\sigma^2)}\,\mathrm{d}A = \int_0^\infty (2\sigma^2 u)^{b/2} e^{-u}\,\mathrm{d}u = (2\sigma^2)^{b/2}\,\Gamma\big(1+\tfrac{b}{2}\big)\,. \tag{4.8}$$

Expected damage. A narrow-band response has one cycle per period, $\nu_0^+ T \approx f_\mathrm{n} T$ cycles in the duration T, each with a Rayleigh amplitude. By (4.3) the expected damage is the number of cycles times $\mathbb{E}[A^b]$:

Gaussian narrow-band damage (Rice–Lalanne)

$$D_G(f_\mathrm{n}) = \frac{K^b}{C}\, f_\mathrm{n}\, T\, \big(2\sigma_z^2\big)^{b/2}\,\Gamma\big(1+\tfrac{b}{2}\big)\,, \qquad \sigma_z^2 = \sum_j W_{ij} G_j. \tag{4.9}$$

Solving for the variance gives the response variance that produces a prescribed damage D in the duration T:

$$\sigma_z^2 = \frac{1}{2}\left(\frac{D\,C}{K^b\, f_\mathrm{n}\, T\,\Gamma(1+b/2)}\right)^{2/b}. \tag{4.10}$$

For wide-band responses (for instance the quasi-static response of a high-frequency oscillator to a strong low-frequency input) the cycles are not one per period and their amplitudes are not Rayleigh distributed. Dirlik's empirical distribution of the rainflow ranges (Appendix C) covers this case; it reduces to (4.9) in the narrow-band limit $\gamma \to 1$. MMVSD uses it (Section 7.5).

4.5 Rainflow on samples: peaks between samples

Rainflow counting works on samples. The true maximum of a cycle generally falls between two samples; the largest sample of the half cycle is the one nearest to it. For a response of frequency f the nearest sample lies within $\pm\Delta t/2$ of the true peak, at a phase offset $\vartheta = 2\pi f\delta$ uniformly distributed in $(-\pi f/f_s, \pi f/f_s)$, and the counted peak is $A\cos\vartheta$ instead of A. A range is measured between a maximum and the following minimum, with independent offsets ϑ_1, ϑ_2; its amplitude is $A(\cos\vartheta_1 + \cos\vartheta_2)/2$, and its damage is reduced on average by the factor

Sampling factor

$$c_b(\rho) = \mathbb{E}\left[\left(\frac{\cos\vartheta_1 + \cos\vartheta_2}{2}\right)^b\right] = \frac{1}{(\pi\rho)^2}\int_0^{\pi\rho}\!\!\int_0^{\pi\rho}\left(\frac{\cos\vartheta_1 + \cos\vartheta_2}{2}\right)^b \mathrm{d}\vartheta_1\,\mathrm{d}\vartheta_2, \qquad \rho = \frac{f}{f_s}, \tag{4.11}$$

(the integrand is even in each angle, so the average over $(-\pi\rho, \pi\rho)$ equals the average over $(0, \pi\rho)$). For $b = 8$: $c_b = 0.98$ at $\rho = 0.04$, 0.86 at $\rho = 0.11$ and 0.54 at $\rho = 0.23$. The

program evaluates (4.11) by Gauss–Legendre quadrature on a table of ρ. Together with the digital response of Section 3.7 this explains why a Gaussian design underestimates the measured damage loss at high natural frequencies when f_s is close to $2.5 f_{\max}$; MMVSD includes both effects in its model.

4.6 Why the kurtosis of the response matters

Let the response be Gaussian given a slowly varying local variance $s(t)$ (Section 2.5). Over a stretch of time where s is constant, the damage rate is the narrow-band rate (4.9) with $\sigma_z^2 = s$. Averaging over time,

$$D = \frac{K^b}{C} f_{\mathrm{n}} T \, \Gamma\big(1 + \tfrac{b}{2}\big) \, 2^{b/2} \, \left\langle s^{b/2} \right\rangle = D_G\big(\langle s \rangle\big) \, \frac{\left\langle s^{b/2} \right\rangle}{\langle s \rangle^{b/2}}, \tag{4.12}$$

where $D_G(\langle s \rangle)$ is the Gaussian damage of a stationary response with the same variance. The factor $\left\langle s^{b/2} \right\rangle / \langle s \rangle^{b/2}$ is at least 1 (Jensen's inequality, since $s^{b/2}$ is convex for $b \geq 2$). For $b = 4$ it equals $\left\langle s^2 \right\rangle / \langle s \rangle^2 = \kappa_z / 3$ by (2.13); for larger b it depends on higher moments of s but grows with κ_z. Two signals with the same PSD can therefore produce very different damage if their responses have different kurtosis, and a test that reproduces the FDS with a different $\kappa_z(f_{\mathrm{n}})$ reproduces it with a different mix of cycles (many moderate cycles instead of a few large ones). This is the reason for prescribing κ_z together with the FDS (Part III).

Part II — Mission synthesis

5 Synthesis of a profile with a prescribed FDS

5.1 The tailoring problem

A target is described by its FDS $D_t(f_{\mathrm{n}})$ over the life T_{L} (Section 4.3), its MRS, its PSD and its statistics. A specification (a PSD or a swept sine) or a mission (Section 6.7) is converted into the same description. The test must last $T \leq T_{\mathrm{L}}$ and reproduce D_t on every node. If the life is compressed by the factor T_{L}/T, the damage per unit time must increase by the same factor; with the narrow-band law (4.9), where $D \propto T \sigma_z^b$, this means multiplying the response amplitudes, hence the test amplitudes, by $(T_{\mathrm{L}}/T)^{1/b}$. For $b = 8$ a ten times shorter test is only $10^{1/8} = 1.33$ times more severe in amplitude; this mild growth is what makes accelerated testing possible.

5.2 Gaussian design: from the FDS to a PSD

The standard mission synthesis looks for a stationary Gaussian profile. Equation (4.10) with the test duration T gives, at every node, the response variance $\sigma_{t,i}^2$ that produces the target damage. Equation (3.10) relates the response variances to the PSD values, so the design is the linear problem

$$\sum_j W_{ij} G_j = \sigma_{t,i}^2, \qquad G_j \geq 0. \tag{5.1}$$

All entries of W are positive and each row is dominated by the columns near its own node (the oscillator responds mainly to its band). The program solves (5.1) by the multiplicative fixed

point

$$G_j \leftarrow G_j \, \frac{\sigma_{t,j}^2}{(WG)_j}, \tag{5.2}$$

started from $G_j = \sigma_{t,j}^2 / W_{jj}$. If the response of node j is too small, $(WG)_j < \sigma_{t,j}^2$ and G_j grows; if too large, it shrinks; positivity is preserved because only positive factors are applied. The iteration stops when $\max_j |\sigma_{t,j}^2/(WG)_j - 1| < 10^{-4}$; when it stalls, a non-negative least-squares solution of the scaled system is used instead. Because neighbouring oscillators overlap, the exact solution may set some nodes to zero and raise their neighbours; the synthesis keeps a tiny floor on such lines so that later multiplicative corrections can revive them.

5.3 Synthesis of the time history

The PSD on the FFT grid of the test (line spacing $1/T$) is the linear interpolation of the nodal values, held constant over $3/Q$ relative bandwidth beyond the first and last node. The time history is the random-phase multisine of Section 2.4 with these line powers. The phases are drawn once with the random seed and then frozen.

5.4 The closed loop

The design uses the Gaussian formula (4.9), while the test will be judged by the rainflow FDS (4.4) of the realised signal; the two differ (wide-band responses, the digital oscillator, the sampling of the peaks, and the scatter of a single realisation). Every FDS method of the package therefore ends with a *closed loop*: the rainflow FDS D_i of the realisation is measured and the spectrum is corrected until it matches the target.

Correction step. If the amplitude of the signal in the band of node i is multiplied by g_i, the response amplitudes of oscillator i are multiplied by g_i and, by (4.3), its damage by g_i^b. The gain that would correct an isolated node is therefore $(D_{t,i}/D_i)^{1/b}$. The program uses

$$g_i = \operatorname{clip}\left(\left(\frac{D_{t,i}}{D_i} \right)^{w/b}, \; \tfrac{1}{1.5}, \; 1.5 \right), \qquad G(f) \leftarrow G(f)\, g(f)^2, \tag{5.3}$$

where $g(f)$ is the linear interpolation of the nodal gains, the phases stay frozen and, for the modulated methods, the envelope is kept. The relaxation w starts at 1 and grows by 25 % after each improving step up to 1.6 (over-relaxation: because neighbouring oscillators share their bands, a plain step corrects only part of the error). A step that makes the result worse is undone and w is halved.

Acceptance score. A step is judged by $\frac{1}{2}\max_i e_i + \frac{1}{2} q_{90}(e)$, where $e_i = |D_i/D_{t,i} - 1|$ and q_{90} is the 90th percentile over the nodes: one stubborn oscillator must not block the improvement of all the others. The loop stops when the maximum error is below the tolerance (5 % by default), when the number of corrections is exhausted, or when $w < 1/16$; the best realisation found is returned.

5.5 Long tests

A test longer than a maximum number of samples (20 million by default; 4 million for MMVSD) is synthesised as a shorter profile of duration T/r that is played r times; the target of the profile is D_t/r. Damage adds over repetitions (Section 4.3), so the whole test meets D_t.

6 The methods of version 2.0

All methods of this section are those of the PhD dissertation, written in a common form. The FDS-controlled methods (`gaussian`, `kfds`, `rf`, `pf`, `psf`, `sor`) end with the closed loop of Section 5.4.

6.1 kFDS: prescribed FDS and kurtosis

Form of the signal. kFDS multiplies a stationary Gaussian carrier g, designed as in Section 5.2, by a slowly varying envelope $e(t)$, (2.10). The envelope is piecewise constant over windows of length L_w, with linear ramps of $L_w/3$ between the levels, so that the waveform has no jumps.

The damage is kept by the normalisation $\langle e^b \rangle = 1$. If a window is long compared with the response time of the oscillators, the response in the window is the carrier's response multiplied by e_w, its cycles are those of the carrier scaled by e_w, and by (4.3) the damage in the window is e_w^b times the carrier's damage in the window. If the carrier's damage is spread evenly over time, summing over the windows gives

$$D(e\,g) = \sum_w e_w^b\, D_{g,w} = \sum_w e_w^b\, \frac{T_w}{T} D_g = \left\langle e^b \right\rangle D_g, \tag{6.1}$$

so any envelope with $\langle e^b \rangle = 1$ keeps the FDS of the carrier. (In the dissertation the same condition appears as weights v_j of blocks, $e_j = (n v_j)^{1/b}$.)

The kurtosis is set by the contrast of the envelope. By (2.13) the kurtosis of the product is $3\langle e^4 \rangle / \langle e^2 \rangle^2$. The envelope is $e_w = \exp(\alpha u_w)$ normalised to $\langle e^b \rangle = 1$, where the relative log-levels $u_w \le 0$ are drawn

- from the distribution of the window RMS of the reference (`pattern = "reference"`, default when a record is available): the logarithm of the RMS of every window of the record is computed, and n_w values are drawn from their life-weighted distribution by quantile resampling, in random order, so that the test has bursts with the statistics of the field data;
- or as random bursts (`"bursts"`): 10 to 20 % of the windows at a high level.

$\alpha = 0$ gives a constant envelope (kurtosis 3); the kurtosis grows with α. The program finds the α that gives the target kurtosis *on the realised signal* $e\,g$ with a bracketing root finder (Brent's method). The envelope is frozen after the first two iterations of the closed loop, which then corrects only the carrier spectrum.

Window length. The normalisation argument requires windows long compared with the response time of the slowest oscillator. The amplitude of a free oscillator decays as $e^{-\zeta\omega_n t}$, with the decay time $1/(\zeta\omega_n) = Q/(\pi f_n) = 2\tau$; the window is at least $2Q/(\pi f_{\min})$. Within this bound the default follows the rule of the dissertation: a fraction of the reference duration (1/150 to 1/400 depending on f_s), with at least 300 samples.

6.2 RF: prescribed FDS and RMS

RF uses the same envelope with the RMS as the target. With $\langle e^b \rangle = 1$ the RMS of the product is $\sigma_g \langle e^2 \rangle^{1/2}$. For $b > 2$ the power-mean inequality gives $\langle e^2 \rangle^{1/2} \le \langle e^b \rangle^{1/b} = 1$, with equality only for a constant envelope; increasing the contrast lowers the RMS. RF can therefore only lower the RMS below that of the Gaussian profile with the same FDS; the program says so when a higher RMS is requested.

6.3 PF and PSF: prescribed FDS and PSD

PF builds the test from two independent stationary Gaussian carriers, a high-energy one x_B (PSD G_B) for the bursts and a low-energy one x_L (PSD G_L) for the rest, combined with a smooth switching mask $m(t) \in [0,1]$ that is 1 during the bursts:

$$x(t) = \sqrt{m(t)}\, x_B(t) + \sqrt{1-m(t)}\, x_L(t). \tag{6.2}$$

Because the carriers are independent, the local PSD is $mG_B + (1-m)G_L$, and the time-averaged PSD is $\langle m\rangle\, G_B + (1-\langle m\rangle)G_L = (t_B G_B + t_L G_L)/T$, with $t_B = \langle m\rangle\, T$ the total burst time and $t_L = T - t_B$. Requiring the reference PSD gives

$$G_L = \frac{T\, G_{\mathrm{ref}} - t_B\, G_B}{t_L}. \tag{6.3}$$

The damage is the sum of the damages of the two parts. The burst PSD is found by a fixed point: start with $D_L = D_t/20$; design G_B for $D_B = D_t - D_L$ over t_B (Section 5.2); compute G_L from the PSD condition and its Gaussian damage D_L over t_L; repeat. The burst time starts at $T/20$ and is halved while G_L would be negative over more than 10 % of the band, but two bursts of at least three decay times of the lowest oscillator are kept when possible. When PSD, RMS and FDS cannot all be met (for instance at high acceleration), the program reports it.

PSF prescribes only the shape of the PSD: PF is run with the damage that the reference produces in the test duration, $D_t\, T/T_{\mathrm{L}}$, and the result is multiplied by $(T_{\mathrm{L}}/T)^{1/b}$, which multiplies its damage by T_{L}/T and its PSD by $(T_{\mathrm{L}}/T)^{2/b}$ without changing the shape.

6.4 Kurtosis control without damage: MLV, VSD, PM

These methods prescribe the PSD and the kurtosis of the signal, not its damage. MLV (multi-level variance) is the modulated Gaussian (2.10) normalised to $\langle e^2\rangle = 1$ (this keeps PSD and RMS) with the contrast solved for the kurtosis. VSD (variable spectral density) splits the band into log-spaced sub-bands, each with its own carrier and independent burst envelope, so that the shape of the PSD changes with time while its time average is the reference PSD. PM (phase manipulation) keeps Gaussian amplitudes and partially aligns the phases of the harmonics in a fraction of the blocks, which creates peaks; the degree of alignment is solved for the kurtosis. The benchmark of the program shows that, over the same duration, their FDS can differ from the reference's by a factor of 3 or more: kurtosis alone does not control the damage.

6.5 MRS control

The largest value of a narrow-band Gaussian response in the duration T can be estimated from Rice's crossing rate. Repeating the derivation of (4.6) for the level $z = u$ instead of 0, the joint density at $z = u$ contains the extra factor $e^{-u^2/(2\sigma_z^2)}$, so the rate of up-crossings of u is $\nu_0^+ e^{-u^2/(2\sigma_z^2)}$. The level crossed on average once in T satisfies $\nu_0^+ T e^{-u^2/(2\sigma_z^2)} = 1$, i.e.

$$u = \sigma_z\sqrt{2\ln(\nu_0^+ T)} \approx \sigma_z\sqrt{2\ln(f_{\mathrm{n}} T)}, \qquad \mathrm{MRS}_G(f_{\mathrm{n}}) = \omega_{\mathrm{n}}^2 \sigma_z \sqrt{2\ln(f_{\mathrm{n}} T)}. \tag{6.4}$$

A target MRS therefore prescribes σ_z at every node, the PSD follows from (5.1), and the closed loop corrects the measured MRS with the exponent 1 instead of $1/b$ in (5.3) (the MRS is proportional to the amplitude).

6.6 Sine on random

Rotating machines add tonal lines to the random vibration. The method `sor` detects the lines in the reference PSD: a peak standing more than a prominence threshold (10 dB by default) above a running median of the PSD is a line, and its amplitude follows from the power in the peak, $A = \sqrt{2P}$ (a sine of amplitude A has the power $A^2/2$). A stationary sine of amplitude A produces in an oscillator a sinusoidal response of amplitude $|H|A$ at the line frequency, i.e. the damage $\frac{K^b}{C} f_0 T (|H|A)^b$ in the time T. To keep the damage of the life T_L in the test duration T, the test amplitude A_T must satisfy $A_T^b T = A^b T_\mathrm{L}$:

$$A_T = A \left(\frac{T_\mathrm{L}}{T} \right)^{1/b}. \tag{6.5}$$

The random part is designed for the remaining damage and corrected in closed loop with the lines kept fixed; nodes where the lines alone exceed the target are reported and left out of the loop criterion. Version 2.1 adds the detection of lines that are present only in some blocks of a record (Section 9).

6.7 Missions

A mission is a tree whose leaves are situations (a record or a specification with the life it represents) and whose inner nodes are groups. In a *series* group the component lives through all children one after the other: the FDS are added, the lives are added, the PSD is the life-weighted mean, the MRS is the maximum. In a *parallel* group it lives through only one of them, not known in advance: the FDS and PSD are enveloped (maximum over the children) and the life, RMS and kurtosis are those of the most damaging child.

The statistics of a series group follow from the moments of a mixture. If a fraction w_c of the life is spent in situation c, where the signal has the moments $\mu_{2,c}$ and $\mu_{4,c}$, the moments over the whole life are the weighted averages $\mu_2 = \sum_c w_c \mu_{2,c}$ and $\mu_4 = \sum_c w_c \mu_{4,c}$, and the kurtosis is μ_4/μ_2^2. Version 2.1 applies the same rule to the response of every oscillator: with $v_c(f_\mathrm{n})$ and $\kappa_c(f_\mathrm{n})$ the variance and kurtosis of the response in situation c, $\mu_{4,c} = \kappa_c v_c^2$ and

Response kurtosis of a series mission

$$v(f_\mathrm{n}) = \sum_c w_c v_c(f_\mathrm{n}), \qquad \kappa_z(f_\mathrm{n}) = \frac{\sum_c w_c \, \kappa_c(f_\mathrm{n}) \, v_c(f_\mathrm{n})^2}{v(f_\mathrm{n})^2}. \tag{6.6}$$

A PSD specification has $\kappa_c = 3$ (it describes a Gaussian process); a swept sine is measured on its generated signal.

Part III — MMVSD

7 The MMVSD model

7.1 Why one envelope is not enough

In kFDS the same envelope multiplies all frequencies. When the windows are long compared with the response times, every oscillator follows the envelope and, by (2.13), every response has the same kurtosis $\kappa_z = 3 \langle e^4 \rangle / \langle e^2 \rangle^2$, whatever f_n. Measured vibration rarely behaves like this.

A vehicle combines a road input whose level changes every few seconds (leptokurtic responses at low f_n), an engine that excites a middle band stationarily (Gaussian responses there) and impacts that excite high frequencies in short bursts (strongly leptokurtic responses at high f_n). The kurtosis of the responses is a function $\kappa_z(f_n)$, and a test that reproduces the FDS with a different function reproduces the damage with a different mix of cycles (Section 4.6).

MMVSD (Moment-Matched Variable Spectral Density) prescribes both functions: the FDS $D_t(f_n)$ and the kurtosis of the responses $\kappa_t(f_n)$, measured on the reference (or combined over a mission with (6.6)). It extends the idea of a variable spectral density, a PSD whose shape changes with time, to a continuous random field designed so that the two families of moments of the responses, their damage and their kurtosis, match the target.

7.2 An evolutionary spectrum

The test signal is described by a *local spectrum* that changes with time:

MMVSD signal model

$$G(f,t) = P(f)\, m(f,t), \qquad m(f,t) \geq 0, \qquad \langle m(f,\cdot)\rangle = 1 \text{ for every } f, \tag{7.1}$$

and, given the field m, the signal is a Gaussian process with this local spectrum.

The time average of the local spectrum is $\langle G(f,\cdot)\rangle = P(f)\,\langle m(f,\cdot)\rangle = P(f)$: the PSD of the test is exactly P, whatever the modulation. The *modulation field* m says when and in which bands the energy is concentrated; its distribution sets how leptokurtic the responses are.

P is parametrised as a smooth correction of a *shape prior* $\Pi(f)$, the PSD of the reference at its full resolution:

$$P(f) = \Pi(f)\, p(f), \qquad p(f) = \sum_j p_j\, \phi_j(f), \tag{7.2}$$

where ϕ_j are the hat functions of the nodes (Section 3.4) extended flat over $3/Q$ beyond the first and last node. The peaks, notches and band edges of the reference are thus kept at full resolution, and the unknowns are the n nodal factors p_j. Outside the band of the grid the test has no energy. The modulation is likewise described by its nodal values $m_j(t)$.

7.3 Local moments of the responses

For a stationary input the spectral moments of the response of oscillator i are (3.8). For the local spectrum (7.1), with P and m written through their nodal values, the moments that the response would have if the local spectrum at time t lasted for ever are

$$S^{\infty}_{k,i}(t) = \sum_j W^{(k)}_{ij}\, p_j\, m_j(t), \qquad W^{(k)}_{ij} = \int f^k\, |H_{d,i}(f)|^2\, |H_{\mathrm{pre}}(f)|^4\, \Pi(f)\, \phi_j(f)\, \mathrm{d}f, \tag{7.3}$$

for $k = 0, 1, 2, 4$. The *moment operators* $W^{(k)}$ generalise Lalanne's operator ($W^{(0)}$ with $\Pi = 1$, the analogue response and no edge extension is the matrix W of (3.10)). MMVSD uses the response H_d of the digital oscillator (3.17) and the pre-filter of Section 3.7, so that the model describes exactly the quantities that the measurement computes. The integrals are evaluated by the trapezoidal rule with 128 intervals per segment between nodes. The model assumes that within the band of an oscillator the local spectrum is the interpolation of its nodal values, i.e. that the modulation varies slowly along the frequency axis compared with the bandwidth f_n/Q.

The response does not follow the local spectrum instantly but with the envelope law (3.13). Applying the low-pass filter F_i of time constant $\tau_i = Q/(2\pi f_{\mathrm{n},i})$ to every moment gives the *local moments*

Local moments of the responses

$$S_{k,i}(t) = F_i\left[\sum_j W_{ij}^{(k)}\, p_j\, m_j(\cdot)\right](t), \qquad F_i[y](t) = \int_0^\infty \frac{1}{\tau_i} e^{-u/\tau_i}\, y(t-u)\,\mathrm{d}u. \tag{7.4}$$

$s_i(t) = S_{0,i}(t)$ is the local variance of the response of oscillator i. On the discrete time grid of the model (step δ, see Section 8.1) the filter becomes the recursion $y_l^F = a_i y_{l-1}^F + (1-a_i)y_l$ with $a_i = e^{-\delta/\tau_i}$, whose gain at zero frequency is $(1-a_i)/(1-a_i) = 1$. The test profile is periodic (it may be played several times), so the recursion is applied circularly, by multiplying the discrete Fourier transform of y by $(1-a_i)/(1-a_i e^{-2\pi \mathrm{j} q/N_t})$ at the frequency index q of the N_t points of the grid.

7.4 Kurtosis of the responses

Given m, the response of oscillator i is Gaussian with the local variance $s_i(t)$. By the last form of (2.13),

Response kurtosis in the model

$$\kappa_i = 3\,\frac{\langle s_i^2\rangle}{\langle s_i\rangle^2}. \tag{7.5}$$

Note that $\langle s_i\rangle = \sum_j W_{ij}^{(0)} p_j \langle m_j\rangle = \sum_j W_{ij}^{(0)} p_j$ does not depend on the modulation (the filter keeps the mean and $\langle m_j\rangle = 1$): the modulation changes the kurtosis and the damage, never the mean response variance.

7.5 Damage of the responses

Over a stretch of time short compared with the variations of s_i but containing many cycles, the response is a stationary Gaussian process with the moments $S_{k,i}(t)$. Its damage rate is the expected damage per second of such a process: Dirlik's rate (Appendix C), a function $d_{\mathrm{Dir}}(S_0,S_1,S_2,S_4)$ that reduces to the narrow-band rate $\nu_0^+(2S_0)^{b/2}\Gamma(1+b/2)$ for narrow bands, multiplied by the sampling factor (4.11) evaluated at the mean crossing rate $\nu_0^+ = \sqrt{S_2/S_0}$ (the cycles that dominate the damage are those of the zero crossings):

Damage in the model

$$D_i = \frac{K^b}{C}\, T\left\langle d\big(S_{0,i},S_{1,i},S_{2,i},S_{4,i}\big)\right\rangle, \qquad d = d_{\mathrm{Dir}}(S_0,S_1,S_2,S_4)\; c_b\!\left(\frac{\sqrt{S_2/S_0}}{f_s}\right). \tag{7.6}$$

For a stationary Gaussian signal ($m \equiv 1$) and a narrow band this is Lalanne's formula (4.9), corrected for the digital oscillator and the sampling. The program also offers the plain narrow-band (Rice) rate instead of Dirlik's.

How well does the model describe a realisation? For a stationary Gaussian profile synthesised by the program at $f_s = 500\,\mathrm{Hz}$ with nodes up to $150\,\mathrm{Hz}$ ($Q = 10$, $b = 8$), the measured response

variances agree with S_0 within 2–7 % at all nodes, and the ratio of the rainflow damage to (7.6) averages about 0.9 (the remaining difference is the scatter of the largest cycles of a 60 s realisation, whose median is below its mean). With the analogue operators and without the sampling factor, the same ratio drops to 0.15 at 113 Hz.

7.6 The modulation field

The field must have, at every node, a prescribed depth of modulation, and between nodes and over time the correlations that describe the bursts of the reference: how long a burst lasts (*burst duration* T_c, possibly different at every node) and how wide a band it covers (*burst width* B, in octaves). It is built in three steps on a grid of K time frames (spacing h_t, the *hop*).

1. A latent Gaussian field. Draw independent standard normal numbers $\xi_{j\ell}$ for every node j and frame ℓ. Smooth every row in time with a Gaussian kernel of standard deviation $T_{c,j}/h_t$ frames (circularly, by multiplying its discrete Fourier transform by $\exp(-2\pi^2\varsigma_j^2\nu^2)$, ν in cycles per frame, $\varsigma_j = T_{c,j}/h_t$), standardise each row to unit variance, then mix the rows with the Gaussian kernel $\exp\!\big(-\tfrac{1}{2}(\log_2 f_{\mathrm{n},j} - \log_2 f_{\mathrm{n},j'})^2/B^2\big)$ along the logarithmic frequency axis, and standardise again. The result $Z_{j\ell}$ is a Gaussian field with zero mean, unit variance at every node, and the prescribed correlations. A Gaussian kernel of standard deviation ς applied to white noise gives the correlation $e^{-\lambda^2/(4\varsigma^2)}$ at the lag λ (the autocorrelation of the kernel is a Gaussian of variance $2\varsigma^2$); the correlation falls to $1/e$ at $\lambda = 2\varsigma$. The settings T_c and B are therefore half the $1/e$ widths of the correlations.

2. Ranks. In every row, replace the values by their ranks $r_{j\ell} \in \{0, \dots, K-1\}$ over time. The ranks keep the information of when the field is high or low, which is all that the copula (the dependence structure) needs.

3. Marginal. The Gamma distribution of shape $a > 0$ and scale $1/a$ has the density $p(m) = a^a m^{a-1} e^{-am}/\Gamma(a)$ for $m > 0$. With the substitution $u = am$, its moments are $\mathbb{E}[m^k] = \frac{1}{a^k\Gamma(a)}\int_0^\infty u^{a+k-1}e^{-u}\mathrm{d}u = \frac{\Gamma(a+k)}{a^k\Gamma(a)}$; since $\Gamma(s+1) = s\Gamma(s)$ (integrate by parts), $\mathbb{E}[m] = \frac{a\Gamma(a)}{a\Gamma(a)} = 1$ and $\mathbb{E}[m^2] = \frac{(a+1)a}{a^2} = 1 + \frac{1}{a}$, so the variance is $1/a$. Map the ranks to the quantiles of this distribution with $a_j = 3/e_j$ (mean 1, variance $e_j/3$):

$$m_{j\ell} = F_{a_j}^{-1}\left(\frac{r_{j\ell} + \frac{1}{2}}{K}\right), \tag{7.7}$$

and divide every row by its mean so that $\langle m_j \rangle = 1$ exactly. Because every row uses the same set of K quantiles, the sample moments of m_j depend only on e_j: there is no sampling noise in $\left\langle m_j^2 \right\rangle$ or $\left\langle m_j^{b/2} \right\rangle$ (a *stratified* copula). The quantiles of the Gamma distribution are tabulated for 200 shapes and interpolated, which makes the field cheap to rebuild during the design.

Meaning of the depth e_j. A Gaussian signal modulated directly by m (local variance proportional to m) has, by (2.13), the kurtosis $3\left\langle m^2 \right\rangle / \left\langle m \right\rangle^2 = 3(1 + \operatorname{Var} m) = 3 + e$; e is the *excess kurtosis of the modulation*. The kurtosis of a response is lower when the bursts are short compared with τ_i, by (3.15); the design (Section 8.4) finds the e_j that give the target κ_t through the full model (7.4)–(7.5), so this reduction is accounted for. The Gamma family is positive, has one shape parameter, contains the exponential distribution ($e = 3$) and tends to the constant 1 as $e \to 0$. The depth is capped (30 by default) to avoid unrealistic peaks.

7.7 Burst width and duration from the record

The program estimates B and $T_{c,j}$ from the reference. The record is cut into frames of 0.5 s (hop 0.25 s, Hann window, i.e. the taper $\sin^2(\pi n/L)$ over the L samples of a frame); in every

frame the power in the 1/3-octave band around every node is computed, and its logarithm $\beta_{j\ell}$ is taken. Even for a stationary signal these values scatter: the power of a band of n_l lines is a sum of $2n_l$ squared Gaussian variables (the real and imaginary parts of each line), i.e. a *chi-square* variable with $\nu = 2n_l$ degrees of freedom up to a scale factor. Half of such a variable, Y, has the Gamma distribution of shape $k = \nu/2$ and scale 1 (density $y^{k-1}e^{-y}/\Gamma(k)$), whose moments are $\mathbb{E}[Y^s] = \int_0^\infty y^{s+k-1}e^{-y}\mathrm{d}y/\Gamma(k) = \Gamma(k+s)/\Gamma(k)$. The logarithm of $\mathbb{E}[e^{s\ln Y}] = \mathbb{E}[Y^s]$, the cumulant generating function of $\ln Y$, is therefore $\ln\Gamma(k+s) - \ln\Gamma(k)$; its second derivative at $s = 0$ is the variance of $\ln Y$, the *trigamma function* $\psi'(k) = \frac{\mathrm{d}^2}{\mathrm{d}k^2}\ln\Gamma(k)$ at $k = \nu/2$ (a scale factor does not change the variance of a logarithm). This noise variance is subtracted from the variance of each band before the correlations are normalised (the effective ν of a band with n_l lines is taken as $2n_l/1.5$, since the Hann window correlates neighbouring lines). Bands whose remaining variance is negligible do not fluctuate and are left out. Then:

- the time correlation of each band is its normalised autocorrelation; the lag where it falls below $1/e$, halved, is $T_{c,j}$ (smoothed along the nodes and interpolated to all of them);
- the frequency correlation is the average correlation between bands as a function of their distance in octaves; the distance where it falls below $1/e$, halved, is B (between 1/12 and 3 octaves).

The burst duration is finally kept above $\max(2\tau_i, 0.25\,\mathrm{s})$: shorter bursts would not reach the responses (Section 3.6) and would only require deeper modulation.

7.8 Keeping the time–frequency pattern of the record

Instead of a random field with the statistics of the record, the field can follow the record itself (`modulation = "pattern"`): the bursts then occur in the same bands and in the same order as in the reference. The logarithm of the band power of the record, $\beta_j(t)$, is computed at every node with the frame length that the synthesis uses at that node (so the time resolution is the one the synthesis can reproduce), smoothed lightly, interpolated on the frame grid of the test and centred, $\tilde\beta_j = \beta_j - \langle\beta_j\rangle$. The field is

$$m_j(t) = \frac{\exp\big(\gamma_j\tilde\beta_j(t)\big)}{\Big\langle\exp(\gamma_j\tilde\beta_j)\Big\rangle}, \tag{7.8}$$

where the exponent $\gamma_j \geq 0$ sets the depth: $\gamma_j = 1$ reproduces the depth of the record, and the design is free to change it to meet κ_t and D_t. The map from γ_j to the excess $e_j = 3\,\mathrm{Var}(m_j)$ is tabulated on a grid of exponents and made monotone, so that the design can work with e_j as for the random field.

A test shorter than the record needs a *mapping* of the reference time onto the test time:

- `blocks` (default): blocks of a few seconds taken evenly across the whole record and concatenated; every part of the record is represented and the burst durations are kept;
- `compress`: the whole record squeezed into the test (the band power is averaged over each test frame); the order of all events is kept and the bursts are shortened;
- a test as long as the record uses it frame by frame (`identity`); a longer one repeats it (`loop`).

The agreement between the band-power pattern of the result and the mapped reference is reported as the median, over the nodes, of their correlation coefficient (1: same pattern).

8 The MMVSD algorithm

8.1 Frames of several lengths

The modulated signal is built from short pieces of noise, *frames*, each with its own spectrum $m(f, t_\ell)$. A frame of length L resolves frequency to $1/L$ and time to about L: long frames are needed at low frequencies (an oscillator needs several cycles in a frame), short ones to follow short bursts. The program therefore uses frames whose length grows towards low frequencies.

Requirements. Node i needs a hop (the spacing of consecutive frames, half the frame length) of at most a third of its burst duration, and frames of at least 8 cycles, i.e. a hop of at least $4/f_{\mathrm{n},i}$. The hop it gets is the larger of the two, $h_i = \max(T_{c,i}/3,\ 4/f_{\mathrm{n},i})$, made non-increasing towards high frequencies (a cumulative minimum from the lowest node up).

Groups. The finest hop $h_0 = \min_i h_i$ (an even number of samples, at least 16) defines the fine time grid. Every node is assigned to a group $g \in \{0, 1, \dots\}$ with the hop $h_0 2^g$: the longest power-of-two multiple of h_0 that does not exceed its need h_i, but long enough for the cycle condition. At most six groups are used, the coarsest with at least 16 frames in the test, and the test length is rounded to a whole number of coarse frames.

Crossovers. Each group synthesises the nodes of its frequency range. Two adjacent groups are joined at a boundary frequency f_b (geometric mean of their edge nodes) with power-complementary weights over 1/6 octave: with

$$t = \min\Big(1,\ \max\Big(0,\ 6\log_2\frac{f}{f_b} + \frac{1}{2}\Big)\Big)$$

the lower group receives $1 - \sin^2(\pi t/2)$ and the upper one $\sin^2(\pi t/2)$ of the power. Applied recursively from the lowest boundary up, this defines weights $X_g(f) \geq 0$ with $\sum_g X_g(f) = 1$ at every frequency.

8.2 Synthesis of one group

Let the group have the hop h (in samples) and the frame length $L = 2h$. Frame ℓ covers the samples $(\ell-1)h, \dots, (\ell+1)h-1$ (circularly) and is multiplied by the *square-root Hann* window $w(n) = \sin\big(\pi(n+\frac{1}{2})/L\big)$, $n = 0, \dots, L-1$. At any sample two frames overlap, at positions n and $n+h$ of their windows, and

$$w(n)^2 + w(n+h)^2 = \sin^2\Big(\frac{\pi(n+\frac{1}{2})}{L}\Big) + \sin^2\Big(\frac{\pi(n+\frac{1}{2})}{L} + \frac{\pi}{2}\Big) = \sin^2(\cdot) + \cos^2(\cdot) = 1, \qquad (8.1)$$

because $\pi h/L = \pi/2$. The squared windows of overlapping frames therefore add up to one.

Each frame is a periodic noise built from its discrete Fourier coefficients on the lines $f_q = q\, f_s/L$, $q = 0, \dots, h$:

$$C_{\ell q} = \sqrt{\tfrac{L f_s}{2}}\ \sqrt{X_g(f_q)}\ \sqrt{m_\ell(f_q)}\ c_{\ell q}, \qquad (8.2)$$

where $c_{\ell q}$ are random coefficients with $\mathbb{E}|c_{\ell q}|^2 = 1$ (Section 8.3) and $m_\ell(f_q)$ is the field of the group (the fine field averaged over the 2^g fine frames of the frame), interpolated linearly in $\log_2 f$ between the nodes. The inverse transform of length L turns the coefficient C of line q into a cosine of amplitude $2|C|/L$, i.e. of power $\frac{1}{2}(2|C|/L)^2 = 2|C|^2/L^2$. With the factor $\sqrt{L f_s/2}$ this power is $(f_s/L)\, X_g m_\ell |c|^2$, and dividing by the line spacing f_s/L gives the PSD $X_g(f)\, m_\ell(f)$ (on average over c): with $m = 1$ and $X_g = 1$ the frame is white noise of unit one-sided PSD.

The frames of a group are independent and have zero mean, so at a given time the variances of the two overlapping (windowed) frames add. Summing over the groups (also independent),

Local spectrum of the synthesised noise

$$m_{\mathrm{eff}}(f,t) = \sum_g X_g(f) \sum_\ell w_{g\ell}(t)^2\, m_{g\ell}(f), \tag{8.3}$$

which is 1 for $m = 1$ (since $\sum_\ell w^2 = 1$ and $\sum_g X_g = 1$) and has time average 1 when every row of m has mean 1. The model evaluates (8.3) on the fine grid with two points per fine hop ($\delta = h_0\Delta t/2$): between two frame centres the weights are $\cos^2(\pi u/2)$ and $\sin^2(\pi u/2)$, $u \in (0,1)$ being the position between them, exactly the squared windows.

Shaping. The sum of the groups, $v(t)$, is a noise whose local spectrum is $m_{\mathrm{eff}}(f,t)$. Its discrete Fourier transform over the whole test, multiplied by $\sqrt{P(f)} = \sqrt{\Pi(f)\,p(f)}$ on every line, gives the test signal: a zero-phase filter that turns the unit spectrum into $P(f)$. The local spectrum of the result is $P(f)\,m_{\mathrm{eff}}(f,t)$, which is the model (7.1) with $m = m_{\mathrm{eff}}$.

8.3 The frame noise

Two kinds of coefficients $c_{\ell q}$ are available.

- *Gaussian frames*: complex normal coefficients (independent real and imaginary parts of variance $\frac{1}{2}$). Every frame is then a Gaussian noise, and so is the sum of frames and its filtered version at every instant, for any frame length: the signal is conditionally Gaussian, as the model assumes (Section 2.4).
- *Multisine frames*: coefficients of modulus 1 with random phases. Each frame has exactly the band power of its spectrum, so the realisation scatters less around the model; but when a frame contains only a few lines in the band of an oscillator ($L\,f_{\mathrm{n}}/Q$ small, typically at low frequency with short frames), the response within the frame is a sum of a few sinusoids whose amplitudes are fixed: its peaks are lighter-tailed than Gaussian, and its damage is on average 20 to 40 % lower than the Gaussian value at the same variance. The closed loop compensates with a higher level and a deeper modulation.

The program uses Gaussian frames (`frame_noise = "gaussian"`), so that the realisations follow the statistics that the model assumes. On measured transport records (not distributed with the package) they gave lower maximum FDS errors and lower κ_z errors after the closed loop than multisine frames; on synthetic records the two were within the scatter between seeds. The scatter itself is a property of the target: with $b = 8$ the damage of a strongly modulated response rests on its few largest bursts, and the damage of a single oscillator typically varies by a factor of about 1.5 between two noises of the same field (standard deviation of about 0.5 in $\ln D$ for a field of depth $e = 3$ in a 60 s test). This is the main reason for the closed loop of Section 8.6.

8.4 Design

The design finds the nodal PSD factors p_j and modulation depths e_j for which the model meets the targets, before any signal is synthesised. The unknowns are taken in logarithms, $y_j = \ln p_j$ and $\chi_j = \ln e_j$, which keeps them positive and makes relative changes uniform. With the field fixed, the model gives the residuals

$$r_{D,i}(y) = \ln \frac{D_i(y)}{D_{t,i}}, \qquad r_{\kappa,i}(\chi) = \kappa_i(\chi) - \kappa_{t,i}. \tag{8.4}$$

Since the kurtosis depends mainly on the depth and the damage on both, the two are solved alternately, three times: first χ for $r_\kappa = 0$ with y fixed, then y for $r_D = 0$ with χ fixed.

Each problem is a regularised least-squares problem solved by the Levenberg–Marquardt method. For the level, for example,

$$\min_{y} \; \|r_D(y)\|^2 + \mu \, \|L(y - y_0)\|^2, \tag{8.5}$$

where L is the second-difference matrix, $(Ly)_i = y_i - 2y_{i+1} + y_{i+2}$, μ the weight of the curvature penalty (0.05 for the level, 0.5 for the depth) and y_0 the preferred solution ($y_0 = 0$ with the shape prior: no correction of the reference shape unless the targets require it). Linearising $r_D(y + \delta) \approx r_D + J\delta$, with the Jacobian $J_{ij} = \partial r_{D,i}/\partial y_j$, and setting the gradient of the objective to zero gives the step

$$\big(J^\top J + \mu L^\top L + \lambda \, \mathrm{diag}(J^\top J + \mu L^\top L)\big)\, \delta = -\big(J^\top r_D + \mu L^\top L (y - y_0)\big). \tag{8.6}$$

The damping λ is divided by 3 after a step that lowers the objective and multiplied by 4 after one that does not (then the step is retried). The depths are limited to $10^{-3} \le e_j \le e_{\max}$. The level design starts from the median level of the stationary Gaussian solution of Section 5.2, the depth design from $e_j = \kappa_{t,j} - 3$.

8.5 Exact Jacobians

The model (7.4)–(7.6) is a composition of linear operators and pointwise functions, so its derivatives can be computed exactly and cheaply.

Damage with respect to the level. On the model grid of N_t points, $D_i = \frac{K^b}{C}\frac{T}{N_t}\sum_l d\big(S_i(t_l)\big)$. Let $g_{k,i}(t) = \partial d/\partial S_k$ evaluated at $S_i(t)$. By (7.4), $\partial S_{k,i}(t)/\partial y_j = W^{(k)}_{ij} F_i[Y_j](t)$ with $Y_j(t) = p_j m_j(t)$ (because $\partial p_j/\partial y_j = p_j$). By the chain rule,

$$\frac{\partial D_i}{\partial y_j} = \frac{K^b}{C}\frac{T}{N_t}\sum_k W^{(k)}_{ij} \sum_l g_{k,i}(t_l)\, F_i[Y_j](t_l). \tag{8.7}$$

The inner sum is a scalar product between $g_{k,i}$ and a filtered signal. For any linear filter F, $\sum_l g(t_l)\, F[Y](t_l) = \sum_l F^\top[g](t_l)\, Y(t_l)$, where the *adjoint* $F^\top$ is the same filter run backwards in time (in the Fourier domain: multiplication by the complex conjugate of its response). Hence

Damage Jacobian

$$\frac{\partial \ln D_i}{\partial y_j} = \frac{1}{D_i}\frac{K^b}{C}\frac{T}{N_t}\sum_k W^{(k)}_{ij} \sum_l F_i^\top[g_{k,i}](t_l)\; p_j\, m_j(t_l). \tag{8.8}$$

The $4n$ adjoint filterings and one matrix product with the field give all n^2 entries at the cost of a few model evaluations. The derivatives $g_{k,i}$ of the closed-form rate d are computed by one-sided differences with a relative step of 10^{-5}.

Kurtosis with respect to the depth. With $A_i = \langle s_i^2 \rangle$ and $B_i = \langle s_i \rangle$, $\kappa_i = 3A_i/B_i^2$. A change $\delta m_j(t)$ of the field row j changes s_i by $\delta s_i = F_i[W^{(0)}_{ij} p_j\, \delta m_j]$. The mean B_i does not change, because the field keeps $\langle m_j \rangle = 1$ and therefore $\langle \delta m_j \rangle = 0$. The mean square changes by $\delta A_i = 2\,\langle s_i\, \delta s_i \rangle = 2 W^{(0)}_{ij} p_j \,\big\langle F_i^\top[s_i]\, \delta m_j \big\rangle$, so

$$\frac{\partial \kappa_i}{\partial \chi_j} = \frac{6\, W^{(0)}_{ij}\, p_j}{B_i^2} \Big\langle F_i^\top[s_i]\, \frac{\partial m_j}{\partial \chi_j} \Big\rangle, \tag{8.9}$$

where $\partial m_j/\partial \chi_j$ is obtained by rebuilding the field with e_j multiplied by $e^{0.05}$ (the field depends smoothly on e_j). The same steps give the derivative of $\ln D_i$ with respect to the depth, used by the closed loop.

Kurtosis with respect to the level. Here B_i changes too: $\partial B_i/\partial y_j = W^{(0)}_{ij} p_j \langle m_j \rangle$ and $\partial A_i/\partial y_j = 2W^{(0)}_{ij} p_j \langle F_i^\top[s_i]\, m_j \rangle$, and $\partial \kappa_i = 3\big(\partial A_i/B_i^2 - 2A_i\,\partial B_i/B_i^3\big)$.

8.6 The closed loop on the realisation

The design is synthesised once with frozen frame coefficients; from then on the signal is a smooth function of (p, e). Every iteration measures, on the realisation, the rainflow FDS D_m and the kurtosis of the responses κ_m (both in one pass over the oscillators) and forms

$$r_{D,i} = \ln \frac{D_{m,i}}{D_{t,i}}, \qquad r_{\kappa,i} = w_\kappa \left(\bar\kappa_{m,i} - \kappa_{t,i}\right), \tag{8.10}$$

with $w_\kappa = 0.15$ and $\bar\kappa_m$ the measured kurtosis averaged over three adjacent nodes. The quality of a realisation is the score $\frac{1}{2}\max_i |r_{D,i}| + \frac{1}{2} q_{90}(|r_D|) + \mathrm{mean}_i |r_{\kappa,i}|$, and the best realisation found is kept. Four kinds of steps are used.

Early re-draws. The first realisation of the design can miss the target damage of one oscillator by a large factor (several hundred per cent) while the median error is a few tens of per cent: with b between 5 and 10, one unusually large burst of the noise within the band of that oscillator carries most of its damage. A correction of the level would then dig a notch into P at that node to compensate one rare event of the noise. Before any correction, therefore, while the worst oscillator misses its damage by more than 50 %, the response z of that oscillator is computed on the realisation, the blocks of its group where $\sum |z|^b$ is largest are found, and the frames that build them (the two largest blocks and their neighbours) receive new random coefficients on the lines within $\pm 1/4$ octave of the oscillator. The new draw is kept only if it lowers both the score and the error of that oscillator; otherwise the old coefficients are restored. At most six draws are made, and the phase ends after two useless draws in a row, because the remaining error is then a property of the design and not of the noise.

Coupled Gauss–Newton steps (first phase of the corrections). The model Jacobians carry the couplings between oscillators (shared bands, the modulation of a neighbouring band, the envelope law), the measured residuals what the model neglects. First the depth moves to correct the kurtosis,

$$\delta\chi = -\big(J_{\kappa\chi}^\top J_{\kappa\chi} + \mu' L^\top L + \lambda\,\mathrm{diag}\big)^{-1}\big(J_{\kappa\chi}^\top r_\kappa + \mu' L^\top L\chi\big), \tag{8.11}$$

then the level moves to correct the damage, including the change that the model predicts for the new depth, $r_D + J_{D\chi}\delta\chi$:

$$\delta y = -\big(J_{Dy}^\top J_{Dy} + \mu'' L^\top L + \lambda\,\mathrm{diag}\big)^{-1}\big(J_{Dy}^\top (r_D + J_{D\chi}\delta\chi) + \mu'' L^\top L(y - y_0)\big), \tag{8.12}$$

with weak curvature weights ($\mu'' = 10^{-4}$, $\mu' = 0.05$) so that node-wise corrections remain possible, and steps limited to ± 0.5 (depth) and ± 0.3 (level) in logarithm. After two steps that do not improve the score, the loop switches to polishing.

Polishing. Only the level moves, by the damped least-squares step

$$\delta y = -w\,\big(J_{Dy}^\top J_{Dy} + \mu_p\,\mathrm{diag}(J_{Dy}^\top J_{Dy})\big)^{-1} J_{Dy}^\top r_D, \qquad \mu_p = 0.5. \tag{8.13}$$

If the Jacobian were diagonal with the entries $b/2$ (each oscillator responding only to its own node, $D \propto p^{b/2}$), this would be $\delta y_i = -w\, r_{D,i}/\big((b/2)(1+\mu_p)\big)$, i.e. the node-wise step (5.3) of version 2.0 ($\ln g_i^2 = (2w/b)\ln(D_t/D)$) reduced by $1+\mu_p$. With the full Jacobian, the correction of an oscillator whose damage comes from a neighbouring band goes to that band instead of driving its own node up or down. The relaxation w adapts as in Section 5.4.

Re-draws. The largest residual errors are usually a property of the realisation: the damage of an oscillator rests on a few large cycles, and the level cannot fix it without disturbing its neighbours.

When the polishing stalls, the frames that carry most of the model damage of the worst oscillator (the two largest and their neighbours, in its group) receive new random coefficients on the lines within $\pm 1/4$ octave of it, and a short branch of four polishing steps follows. The branch is kept only if it beats the best realisation so far; otherwise the old coefficients are restored. At most eight re-draws are tried.

Stopping. The loop stops when the maximum FDS error is below the tolerance and the median kurtosis error below 0.3, when the iteration budget is used, or when the score has not improved by 1 % for ten iterations.

A second strategy. The coupled loop corrects kurtosis and damage together and usually converges in a few tens of iterations. Where the damage of a few oscillators rests on a few bursts it can stall at a compromise between neighbouring oscillators, because the model Jacobian then describes the realisation poorly. On request (option `fallback`), when its maximum FDS error ends above twice the tolerance, the program runs a second closed loop from the same design and the same noise with node-wise steps only (the step of Section 5.4, the depth kept at the design), which progresses more slowly but steadily, and keeps the realisation with the better score. The option is off by default: on measured transport records the coupled loop was better in every case, and the second loop only doubled the computing time. The program warns when the final level differs from the design by more than a factor of 8 over a band (a systematic drift, typically at oscillators whose damage is produced by another band), and when the target kurtosis could not be reached with the maximum depth.

9 Sine-on-random blocks

9.1 The problem

Rotating machines (engines, pumps, fans) add tonal lines to the random vibration, at the rotation frequency and its orders. In a record these lines are often present only in some parts (idle, cruise) and their frequency drifts with the speed. A line is not a narrow random resonance: its phase is coherent over many periods, its amplitude is nearly constant, and a sinusoidal response is far less damaging than a Gaussian response of the same variance (a sine has no rare large cycles: its damage ratio to a narrow-band Gaussian response of equal variance is $1/\Gamma(1 + b/2)$, which is $1/24$ for $b = 8$, from (4.8) with $A = \sqrt{2}\sigma$ fixed). Treating the lines as part of a random process therefore misrepresents the damage. MMVSD detects the blocks of the record where lines stand on the random background (*sine-on-random blocks*), reproduces the lines as deterministic sines, and designs the random part for the rest.

9.2 Detection block by block

The record is cut into blocks of 5 s. In every block the PSD is estimated with a resolution of 0.5 Hz (Hann windows, 75 % overlap) and the line detector of Section 6.6 proposes candidate peaks. A candidate is accepted only if it passes a *phase coherence* test.

Coherence. The frequency of the candidate is followed frame by frame inside the block (the maximum of each short-time spectrum near the candidate, refined by fitting a parabola to the logarithm of the three largest values), and a straight line $f(t)$ is fitted to these values. The block is then demodulated along this frequency, $y(t) = x(t)\, e^{-2\pi \mathrm{j} \int f \,\mathrm{d}t}$, and low-pass filtered to ± 1 Hz, which leaves the *complex envelope* $c(t)$ of whatever lies near the candidate. For a sine $A\cos(\varphi(t) + \psi)$ with $\varphi = 2\pi \int f \mathrm{d}t$,

$$x\, e^{-\mathrm{j}\varphi} = \tfrac{A}{2} e^{\mathrm{j}\psi} + \tfrac{A}{2} e^{-\mathrm{j}\psi} e^{-2\mathrm{j}\varphi}, \tag{9.1}$$

and the low-pass filter removes the second term (at twice the line frequency): $c = \frac{A}{2} e^{\mathrm{j}\psi}$ is constant. For narrow-band noise $c(t)$ is a random phasor. With N samples of c, the coherence is

$$\Phi = \max_{|\delta| \le r/4} \frac{\left| \sum_l c_l \, e^{-2\pi \mathrm{j} \delta t_l} \right|^2}{N \sum_l |c_l|^2} \in [0, 1], \tag{9.2}$$

where the small residual frequency offset δ (up to a quarter of the resolution r) absorbs the error of the frequency fit. By the Cauchy–Schwarz inequality $|\sum_l c_l|^2 \le N \sum_l |c_l|^2$, with equality when c is constant: a sine gives $\Phi \approx 1$ (slightly less because of the background in the filter band). Noise of bandwidth B observed for the block duration T_b has a complex envelope that stays correlated for about $1/B$; the N samples form about $N_{\mathrm{ind}} = BT_b$ independent groups of N/N_{ind} nearly equal values. Each group sums to about $(N/N_{\mathrm{ind}})c_g$, the groups add with random phases, so $\mathbb{E}|\sum_l c_l|^2 \approx N_{\mathrm{ind}}(N/N_{\mathrm{ind}})^2 \mathbb{E}|c|^2 = N^2 \mathbb{E}|c|^2 / N_{\mathrm{ind}}$, while $N \sum |c|^2 \approx N^2 \mathbb{E}|c|^2$: noise gives $\Phi \approx 1/(BT_b)$. A random resonance with $Q = 60$ at 70 Hz ($B \approx 1.2$ Hz) over 5 s gives $\Phi \approx 0.17$; a line gives 0.9 or more.

Tracks. A candidate with $\Phi \ge 0.75$ is a line; candidates of consecutive blocks within 5 % in frequency are linked into *tracks*. A track is kept if at least two of its blocks are strongly coherent ($\Phi \ge 0.75$); between them it may pass through weakly coherent blocks ($\Phi \ge 0.5$), so that a line with a wandering speed or a low signal-to-noise ratio is not cut (hysteresis). A random resonance that passes the test by chance in one block is dropped.

9.3 Extraction

Each track is followed frame by frame in its active intervals (consecutive detected blocks, gaps of up to two blocks bridged), its frequency is median-filtered over three frames against spurious peaks, and the record is demodulated along the track's own phase and low-pass filtered (to half the resolution). By the computation above, the low-pass result is the complex envelope $c(t)$ of the line, and the line itself is

$$\ell(t) = 2 \operatorname{Re}\bigl(c(t) \, e^{\mathrm{j}\varphi(t)}\bigr), \tag{9.3}$$

tapered at the edges of the intervals. The sum of the lines is subtracted from the record; the *residual* is the random part of the record. The track keeps its frequency $f(t)$ and amplitude $a(t) = 2|c(t)|$ on a time grid of the record.

9.4 The lines of the test

The reference time is mapped onto the test time with the mapping of the time–frequency pattern (Section 7.8); with the random field, `compress` is used, so that every tonal block of the record appears in the test, in its order. The amplitude and frequency of each track are read along the mapped time, smoothed over 0.1 s, and the track is played as a sine with a random initial phase.

Gain of a track. The damage that a line with a slowly varying amplitude $a(t)$ produces in an oscillator is the sum of the damages of its cycles, $\frac{K^b}{C} \int f \, (|H|a)^b \, \mathrm{d}t$. If the test plays the same frequency path, the factors $f|H|^b$ are the same in the record and in the test, and the life damage is kept when the test amplitude $g_j a_j$ satisfies $\int_{\mathrm{test}} (g_j a_j)^b \mathrm{d}t = \frac{T_{\mathrm{L}}}{T_{\mathrm{ref}}} \int_{\mathrm{ref}} a_j^b \mathrm{d}t$:

Gain of a track

$$g_j = \left(\frac{T_{\mathrm{L}}}{T_{\mathrm{ref}}} \, \frac{\int_{\mathrm{ref}} a_j(t)^b \, \mathrm{d}t}{\int_{\mathrm{test}} a_j(t)^b \, \mathrm{d}t} \right)^{1/b} . \tag{9.4}$$

For a stationary line, $\int_{\text{ref}} a^b = A^b T_{\text{ref}}$ and $\int_{\text{test}} a^b = A^b T$, and $g = (T_{\text{L}}/T)^{1/b}$, the sine-on-random rule (6.5). A track whose blocks fall outside the mapped part of the record is reported; its damage is then carried by the random part.

9.5 The random part

Line damage and random damage do not add: a line rides on the random peaks and changes their ranges. Rather than assuming additivity, MMVSD designs the random part for the *measured* FDS and response kurtosis of the residual, scaled to the life, $D_{\text{res}} T_{\text{L}}/T_{\text{ref}}$ and $\max(\kappa_{\text{res}}, 3)$, with the PSD of the residual as shape prior and, in pattern mode, the band-power pattern of the residual. The closed loop then measures the complete test (random part plus lines) against the complete target, with the lines kept fixed.

Kurtosis of a response with lines. Given the field, the response of an oscillator is a Gaussian part g (variance s) plus the responses of the lines, $\sum_j u_j^{1/2}\sqrt{2}\cos\theta_j$, where u_j is the response variance of line j and the phases θ_j are independent and uniformly distributed. For one line of amplitude $\alpha = \sqrt{2u}$: $\mathbb{E}[\alpha^4 \cos^4\theta] = \alpha^4 \cdot \frac{3}{8} = \frac{3}{8}(2u)^2 = \frac{3}{2}u^2$, using $\cos^4\theta = \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$. For the sum $X = \sum_j X_j$ of independent zero-mean lines, the fourth moment keeps the terms X_j^4 and the pairs $X_j^2 X_l^2$ ($\binom{4}{2} = 6$ orderings, counted once for each unordered pair, i.e. 3 per ordered pair $j \neq l$):

$$\mathbb{E}[X^4] = \sum_j \tfrac{3}{2}u_j^2 + 3\sum_{j\neq l} u_j u_l = 3U^2 - \tfrac{3}{2}\sum_j u_j^2, \qquad U = \sum_j u_j, \tag{9.5}$$

because $3\sum_{j\neq l} u_j u_l = 3(U^2 - \sum_j u_j^2)$. Adding the independent Gaussian part, whose odd moments vanish, $\mathbb{E}[(g+X)^4] = \mathbb{E}g^4 + 6\mathbb{E}g^2\mathbb{E}X^2 + \mathbb{E}X^4 = 3s^2 + 6sU + 3U^2 - \frac{3}{2}\sum u_j^2$:

Fourth moment of a Gaussian response with lines

$$\mathbb{E}[z^4 \mid m] = 3(s+U)^2 - \tfrac{3}{2}\sum_j u_j^2, \qquad \kappa = \frac{3\left\langle (s+U)^2 \right\rangle - \frac{3}{2}\left\langle \sum_j u_j^2 \right\rangle}{\left\langle s+U \right\rangle^2}. \tag{9.6}$$

The line variances $u_j(t) = \frac{1}{2}a_j(t)^2 |H_d(f_j(t))|^2 |H_{\text{pre}}|^4$ are filtered by the envelope law like s. The kurtosis model of the closed loop uses this formula; the damage of the lines has no reliable closed form on a broad-band random response, so it is not modelled but measured.

Damage elasticity. Without lines, multiplying the PSD by λ multiplies the amplitudes by $\sqrt{\lambda}$ and the damage by $\lambda^{b/2}$: $\partial \ln D / \partial \ln p = b/2$. With lines part of the damage is fixed and the elasticity η_i is lower. It is measured on the first realisation by amplifying the random part by 1.1 (its PSD by 1.1^2):

$$\eta_i = \frac{\ln\big(D_i(1.1\,x_r + x_\ell)/D_i(x_r + x_\ell)\big)}{2\ln 1.1}, \tag{9.7}$$

limited to $[0.1\,b/2,\ b/2]$, and the damage rows of the closed-loop Jacobians are multiplied by $\eta_i/(b/2)$. Where the lines alone exceed the target damage, the program reports that those frequencies are over-tested, as the sine-on-random method of version 2.0 does.

10 From the theory to the settings

10.1 The damage model

The damage model (natural frequency grid, Q, b, C, K, pre-filter) is shared by all computations of a workspace. The FDS of a record is computed by default with the exact procedure of

Section 4.3 (`time`); `frequency` uses the Gaussian formula (4.9) on the Welch PSD, `fast` a self-calibrated approximation. If $f_{\max} > f_s/2.5$ the record is first up-sampled by 5/4, so that every oscillator has at least 2.5 samples per period. The exact computation also returns the variance and the kurtosis of every response, $\kappa_z(f_n)$, in the same pass; the program computes the oscillators in parallel (worker processes for long records), and the results do not depend on the number of workers.

10.2 Choosing a method

Situation	Method
Only the damage matters; a Gaussian test is acceptable	`gaussian`
Leptokurtic record, one kurtosis for the whole signal	`kfds`
The test must keep the RMS of the field	`rf`
The test must keep the PSD (shape) of the field	`pf`, `psf`
Stationary tonal lines over the whole record	`sor`
Kurtosis of the responses different in different bands; lines in some blocks only; bursts at the right time	`mmvsd`
Peak severity (shock-like environments)	`mrs`
Kurtosis control without damage equivalence	`mlv`, `vsd`, `pm`
The S–N exponent is uncertain, the test must be right for every b; a record is available and the compression is modest (Section 12)	`edit`

10.3 MMVSD settings

Interface	Command line	Meaning
Modulation	`--modulation`	`random`: the field of Section 7.6 with the burst statistics of the target; `pattern`: the time–frequency pattern of the record (Section 7.8)
Shorter test: mapping	`--mapping`	`blocks` or `compress` (Section 7.8)
Block length	`--block`	length of the blocks of the `blocks` mapping (default: 4 s or five burst durations)
Sine-on-random blocks	`--lines`	`lines`: detected lines as deterministic sines (Section 9); `random`: detected but left in the random part; `off`
Burst width / duration	`--freq-corr`, `--time-corr`	B in octaves and T_c in seconds (Section 7.7); estimated from the record by default
Max. modulation excess	`--max-excess`	cap $e_{\max}$ of the depth (default 30)
Design anchor	`--anchor`	pull of the level towards the design at the oscillators whose damage comes from another band (default 0)
Max. corrections, tolerance	`--iterations`, `--tol`	budget and stopping tolerance of the closed loop
Seed	`--seed`	the random numbers of the field and of the frames: the same seed gives the same profile

For a mission or a specification there is no single record: the target κ_z is the mixture of Section 6.7, the burst statistics are estimated on the longest situation, and only the random field is available. Lines are detected on single records only.

10.4 Edit settings

Interface	Command line	Meaning
Window	`--window`	window length L_w of Section 13.1 (default: 2 s or three amplitude decay times of the slowest oscillator)
Window ranking	`--ranking`	`common` (sum of the damage shares) or `fair` (max–min over the oscillators), Section 13.2
Ring-down margin	`--margin`	extension of every kept window, in amplitude decay times $Q/(\pi f_{\min})$ (default 2)
Cross-fade	`--fade`	raised-cosine fade between kept stretches [s] (default 0.1)
Spectral gain	`--no-gain`	off: editing only, the FDS stays below the target beyond the limit; on (default): Section 13.5
Gain cap	`--gain-max`	clip of the gain and of its inverse (default 2)
Retention for the limit	`--retention`	share ρ of (12.4) that defines the reported compression limit (default 0.9)
Max. corrections, tolerance	`--iterations`, `--tol`	budget and stopping tolerance of the gain loop
Compressed kurtosis target	`--compress-kurtosis`	for kFDS, RF, MLV, VSD, PM: the kurtosis of the record edited to the test duration replaces the record's kurtosis as target (Section 14)

`edit` needs a record (or a result) as target; a mission or a specification has no time history to edit. The analysis tab computes the Lorenz curve and the compression limits of a record without building a test ("Compression limit"), and the command line with `analyze --compressibility`.

10.5 Reading the results

- *FDS error*: the ratio of the rainflow FDS of the profile (times its repetitions) to the target, node by node; the maximum and the median of $|\text{ratio} - 1|$ and the share of nodes within 10 % are reported. The maximum is governed by the few oscillators whose damage rests on a few large cycles (low natural frequencies, strong modulation); the median shows the overall agreement.
- κ_z *error*: the median of $|\kappa_{z,\text{profile}} - \kappa_{z,\text{target}}|$ over the nodes, computed for every method (it shows, for instance, that a Gaussian test has $\kappa_z \approx 3$ everywhere).
- *MRS ratio*: above 1 the test produces larger peak responses than the target; check it against the strength of the component.
- *PSD deviation*: the RMS, over the band, of the ratio in dB between the PSD of the profile and that of the target.
- *Pattern correlation* (pattern mode): Section 7.8.
- *Compression limit and spectral gain* (`edit`): the requested compression against the limit (12.4), the Lorenz curves of the record (mean and worst oscillator, with the lognormal closed form and the diagonal of a stationary signal), the stretches kept, and the gain applied at every node. A gain of 1 everywhere means that editing alone met the

target and the test is right for every exponent; a gain above 1 in a band means that the amplitude did the work there and the test depends on b in that band (Section 13.5). The sensitivity card (next section) shows the dependence directly.

- *Warnings*: a level drifted far from the design over a band, a kurtosis target that cannot be reached with the maximum depth, bursts shorter than two response times (filtered by the oscillators, Section 3.6), lines that exceed the target damage alone or that are not in the mapped part of the record; for `edit`, a compression beyond the limit and a gain that reached its cap.

10.6 Limits

The FDS and the response kurtosis describe the damage of linear, lightly damped oscillators with a Basquin law; they are only as good as the chosen Q and b (the program computes the sensitivity of a result to other values). The kurtosis of the responses is a fourth-moment description: two signals with the same κ_z can still differ in their largest peaks, which the MRS ratio shows. The model of Section 7 assumes that the modulation varies slowly along frequency compared with the oscillator bandwidth; very narrow bursts in frequency are smoothed. Finally, a test that is much shorter than the life compresses the bursts: their duration is kept in the `blocks` mapping, at the price of leaving out parts of the record. Part IV makes this last point quantitative: the Lorenz curve of the record says how much can be left out before the amplitude has to rise, and with it the dependence on the exponent.

11 Diagnostics and checks

The interface offers a set of diagnostics of a single signal and of a result. This section defines them.

11.1 Statistics over time and amplitude

Running RMS and kurtosis. The signal is cut into consecutive windows of L_w seconds; in each window the RMS and the kurtosis (2.2) are computed. A Gaussian stationary signal gives a nearly constant RMS and a kurtosis near 3 in every window; bursts show up as RMS peaks. The distribution of the 1-second RMS over the record (shown for a result and its target) describes the burst structure that kFDS and MMVSD reproduce.

Probability density. The histogram of the samples is compared with the Gaussian density with the same mean and standard deviation, on a logarithmic scale where the tails are visible.

11.2 Time–frequency diagnostics

Spectrogram. The short-time Fourier transform (STFT) cuts the signal into overlapping windowed segments (Hann window, 50 % overlap) and computes the PSD of each, giving the power as a function of time and frequency.

Spectral kurtosis. Let $X(f, t)$ be the STFT coefficient at frequency f of the segment at time t. The spectral kurtosis is

$$SK(f) = \frac{\left\langle |X(f,\cdot)|^4 \right\rangle}{\left\langle |X(f,\cdot)|^2 \right\rangle^2} - 2. \tag{11.1}$$

For a stationary Gaussian signal, $X = U + \mathrm{j}V$ with U and V independent zero-mean Gaussian variables of equal variance $\sigma^2/2$. Then $|X|^2 = U^2 + V^2$ has the exponential distribution with mean σ^2 (the squared length of a two-dimensional Gaussian vector; with Section 4.4, $p(A) = (2A/\sigma^2)e^{-A^2/\sigma^2}$ for $A = |X|$, and the substitution $u = A^2$ gives $p(u) = e^{-u/\sigma^2}/\sigma^2$), whose second moment is $\int_0^\infty u^2 e^{-u/\sigma^2}\mathrm{d}u/\sigma^2 = 2\sigma^4$. Hence $SK = 2\sigma^4/\sigma^4 - 2 = 0$. A band whose power changes from segment to segment (bursts) has $SK > 0$; a stationary sine, whose $|X|$ is constant, has $SK = 1 - 2 = -1$. The *kurtogram* shows SK as a function of frequency and of the window length, since short bursts are seen only with short windows.

Response of one oscillator. The pseudo-acceleration $\omega_\mathrm{n}^2 z(t)$ of a chosen oscillator, with its statistics, shows directly how the input is filtered (Section 3.6). The curves of the kurtosis and standard deviation of all responses versus f_n are the quantities $\kappa_z(f_\mathrm{n})$ and $\sigma_{\omega_\mathrm{n}^2 z}(f_\mathrm{n})$ of Part I, computed in one pass over the oscillators.

11.3 Approximate FDS

Besides the exact rainflow FDS and the Gaussian formula, the program offers the fast approximation of the dissertation. For a narrow-band response, the large values of z occur near the peaks of the cycles, and a cycle of amplitude A contributes to $\sum_k z_k^b \Delta t$ (over the samples where z is positive) an amount proportional to A^b/f_n. Hence

$$D(f_\mathrm{n}) \approx h(f_\mathrm{n}) \, \frac{K^b}{C} \, f_\mathrm{n} \, \Delta t \sum_{k:\ z_k > \rho\, z_\mathrm{max}} z_k^b, \qquad \rho = 400^{-1/b}, \tag{11.2}$$

where only the samples above the fraction ρ of the maximum are summed (smaller cycles contribute little to the damage when b is large) and $h(f_\mathrm{n})$ absorbs the proportionality constant and the residual effects of the cycle shapes. The program calibrates $h(f_\mathrm{n})$ on the first 20 s of the signal itself against the exact rainflow damage.

11.4 Sensitivity to b and Q

A profile is designed with one damage model. If the true exponent of the S–N curve or the damping of the component differ, the profile and the target no longer have the same FDS. The program recomputes the FDS of both with other values of b or Q and shows their ratio: a ratio that stays close to 1 means that the equivalence does not depend on the exact damage model.

For an `edit` result the sensitivity card is the direct check of Section 13.5: the FDS error at other exponents stays small where the gain is 1 and grows as $g^{\,b'-b}$ where the gain acted.

11.5 Compression limit

The analysis tab computes, for a record, the damage Lorenz curves of Section 12.3 (mean over the oscillators and worst oscillator) and the compression limits (12.4) for 80, 90 and 95 % of the damage, together with the lognormal closed form (12.7) for the record's kurtosis. The window defaults to that of the `edit` method. The curve is read before choosing a test duration: at a compression within the limit the `edit` method gives a test that is right for every exponent; beyond it every method, `edit` included, relies on amplitude, and the choice between them is the choice of what else to prescribe (Section 10).

11.6 Matching the amplitude distribution

A profile can be given exactly the amplitude distribution of the target by a monotone mapping of its samples: the sample of rank r (out of N) is replaced by the quantile of the target values at the probability $(r+\frac{1}{2})/N$. The order of the samples, hence the time structure, is kept; the PSD and the FDS change slightly, so the program measures them again after the mapping.

11.7 Shaker limits

A shaker has limits of acceleration, velocity and stroke (peak-to-peak displacement). Velocity and displacement are obtained from the acceleration in the frequency domain, dividing every line by $\mathrm{j}\omega$ (velocity) or by $-\omega^2$ (displacement), after a high-pass filter with the gain $1/\sqrt{1+(f_{\mathrm{hp}}/f)^8}$ (the magnitude of a fourth-order Butterworth filter): without it the lowest lines, divided by a very small ω^2, would dominate the displacement. If the stroke is exceeded, the program searches the lowest high-pass frequency that brings it within the limit; applying that filter in the pre-processing changes the profile only below that frequency.

Part IV — The compression limit and the edit method (version 2.2)

12 The compression limit

Every method of Parts II and III reproduces the FDS at one Basquin exponent b, the one of the damage model. This section asks what happens at the other exponents, and arrives at a statement about how far a record can be shortened at all. The answer is a curve that can be computed from the record alone, before any synthesis.

12.1 Damage as a moment of the cycle histogram

Fix one oscillator of the grid and collect the rainflow cycles of its response: N cycles with amplitudes A_c (half cycles count with $n_c=\frac{1}{2}$). Write their distribution as a density $p(A)$, so that $Np(A)\,\mathrm{d}A$ is the number of cycles with amplitude in $(A, A+\mathrm{d}A)$. The damage (4.3) is then

$$D_b = \frac{K^b}{C}\sum_c n_c A_c^b = \frac{K^b}{C}\,N\int_0^\infty A^b\,p(A)\,\mathrm{d}A = \frac{K^b}{C}\,N\,\mathbb{E}[A^b]. \tag{12.1}$$

The damage at exponent b is the b-th moment of the cycle-amplitude distribution (in the language of integral transforms, the Mellin transform of p at $b+1$). The FDS of the damage model is one such moment per oscillator. Two consequences follow at once.

The FDS at every exponent is the cycle histogram

If two signals have the same rainflow cycle distribution at an oscillator (same N, same p), their damage is equal for every exponent b. Conversely, if their damage is equal for every b in an interval, the distributions coincide (the moments of a distribution with a light tail, such as the Rayleigh mixtures of Section 4.4, determine it). "Equal FDS for every b" and "equal cycle histogram" are the same statement.

This is the reason to care about other exponents: the exponent of a real component is uncertain (material, weld class and surface change it), and a test that is right for every b does not depend on that uncertainty. It is also a description independent of Miner's rule, usable with any damage model that is a function of the cycle amplitudes.

12.2 Why a shorter test cannot be exact for every exponent

Let a test profile be built from the record with an amplitude factor λ and the duration T/α. Let N_t and N_r be the numbers of cycles of the test and of the record at an oscillator, and $\mathbb{E}_t$, $\mathbb{E}_r$ the averages over their cycles. Equal damage for every b requires

$$\lambda^b \, N_t \, \mathbb{E}_t[A^b] = N_r \, \mathbb{E}_r[A^b] \qquad \text{for every } b. \tag{12.2}$$

Set $b = 0$: $\lambda^0 = 1$ and $\mathbb{E}[A^0] = 1$, so $N_t = N_r$. But the cycles of a narrow-band response arrive at the rate of the natural frequency (one per period, Section 4.4), so a test of duration T/α holds about N_r/α cycles. The two statements are compatible only for $\alpha = 1$.

No scaling is exact for every exponent

A profile obtained by scaling a shorter signal cannot reproduce the damage of the record at every exponent, because the moment of order zero, the number of cycles, is reduced by the compression.

The way out is in the same equation. The small cycles make up most of the count, but at a large exponent they contribute almost nothing to the damage. If a shorter test drops exactly the small cycles and keeps the large ones, the number of cycles changes (the zero-order moment is lost) while the high moments are preserved. On a time history there is one operation that does this: keep the stretches of the record that carry the damage, unchanged, and leave out the quiet ones. Section 13 builds a method on it. How far it can go is the question of the next subsection.

The error of dropping the cycles below an amplitude A_0 is the share of the damage they carry,

$$\varepsilon_b = \frac{\int_0^{A_0} A^b p(A)\,\mathrm{d}A}{\int_0^{\infty} A^b p(A)\,\mathrm{d}A}, \tag{12.3}$$

which decreases with b. For a Rayleigh distribution and $A_0 = \sigma_z$ (which discards 39 % of the cycles), the substitution of (4.8) gives the regularised incomplete Gamma function $P(1+b/2,\, \frac{1}{2})$: $\varepsilon_3 = 0.037$, $\varepsilon_5 = 0.005$, $\varepsilon_9 = 0.00005$. A large exponent does not notice the loss; a small one does, which is why an edited test is below the record at $b = 3$ in the tables of Section 13.6.

12.3 The damage Lorenz curve

Cut the record into consecutive windows of length L_w (Section 13.1 says how L_w is chosen) and compute, for every oscillator i of the grid, the share W_{ij} of its damage that falls in window j: the rainflow damage of the response in that window, divided by the damage over the whole record. Every row of W sums to 1. Sort the windows from the most to the least valuable (the ranking is defined in Section 13.2) and add up the shares along the ranking.

Damage Lorenz curve and compression limit

Let q be the fraction of the time kept (the first qM windows of the ranking, out of M) and $L_i(q)$ the share of the damage of oscillator i contained in them. L_i is the damage Lorenz curve of the oscillator: $L_i(0) = 0$, $L_i(1) = 1$, non-decreasing and concave, and equal to the diagonal $L(q) = q$ when the damage is spread uniformly over the time. The curve of the worst-served oscillator, $L_{\min}(q) = \min_i L_i(q)$, sets the compression limit: the largest time compression at which editing alone keeps a share ρ of the damage at every oscillator is

$$\alpha_{\max}(\rho) = \frac{1}{q(\rho)}, \qquad L_{\min}\big(q(\rho)\big) = \rho. \tag{12.4}$$

Three properties follow from the definition. The curve lies higher the more the damage is concentrated in time: a stationary Gaussian record has the diagonal and cannot be shortened by editing at all, a record with bursts can. It also rises with b, because a larger exponent concentrates the damage in fewer cycles: a large exponent allows more compression. And it is different for every oscillator: the damage of a low-frequency oscillator excited by a few shocks sits in a few windows, that of an oscillator driven by a stationary band is spread out. A test has to be right at every natural frequency, so the binding curve is $L_{\min}$, not the mean over the oscillators (which the interface shows as well, because it tells how much is lost on average).

Beyond $\alpha_{\max}$ the record has no more quiet time to drop. The remaining compression can only come from raising the amplitude, and with it the dependence on the exponent returns, because an amplitude factor acts as its b-th power on the damage. The two mechanisms are therefore not competitors but two stages: editing up to the limit, amplitude beyond it, with the dependence on b confined to the part the amplitude did. A test specification is described by the pair, not by α alone.

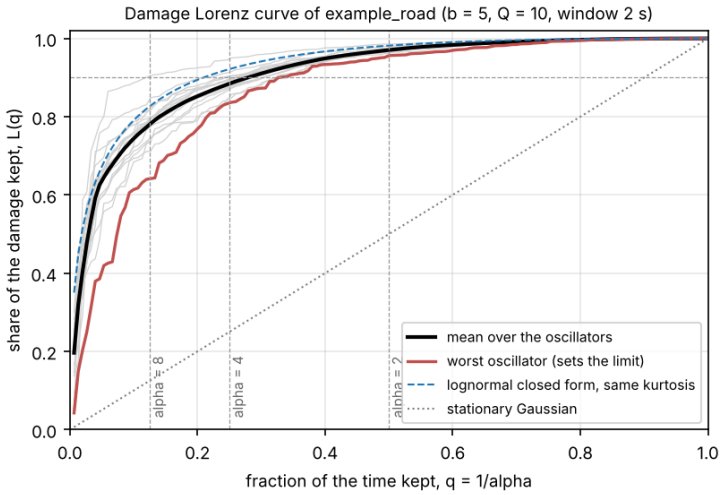


Figure 1: Damage Lorenz curves of the example record (grey: one curve per oscillator). Keeping the most damaging quarter of the time keeps 88 % of the damage on average and 82 % at the worst oscillator; the 90 % limit is $\alpha_{\max} = 3.0$. The dashed curve is the closed form of Section 12.4 for the kurtosis of the record.

12.4 Closed form for a lognormal envelope

The Lorenz curve of a modulated Gaussian signal $x = e\,g$ (Section 2.5) can be written in closed form when the local variance $s = e^2\sigma_g^2$ is lognormal. The result is useful as an estimate before

any computation and as a reference against which a measured curve is compared.

Let $s = e^Y$ with Y Gaussian of mean μ and standard deviation τ. By Section 4.6 the damage of a stretch of time with local variance s is proportional to $s^{b/2}$, so the damage share of the stretches with $Y > y_0$ is $\mathbb{E}[s^{b/2}\mathbf{1}_{Y>y_0}]/\mathbb{E}[s^{b/2}]$, and their time share is $q = P(Y > y_0)$. Two facts about Gaussian variables are needed. First, $P(Y > y_0) = \Phi(u)$ with $u = (\mu - y_0)/\tau$ and Φ the standard normal distribution function. Second, the truncated expectation of an exponential of Y: writing $c = b/2$,

$$\mathbb{E}\big[e^{cY}\mathbf{1}_{Y>y_0}\big] = \int_{y_0}^{\infty} e^{cy}\,\frac{1}{\tau\sqrt{2\pi}}e^{-(y-\mu)^2/(2\tau^2)}\,\mathrm{d}y. \tag{12.5}$$

Completing the square in the exponent, $cy - \frac{(y-\mu)^2}{2\tau^2} = c\mu + \frac{c^2\tau^2}{2} - \frac{(y-\mu-c\tau^2)^2}{2\tau^2}$, the integrand becomes a Gaussian density centred at $\mu + c\tau^2$ times the constant $e^{c\mu+c^2\tau^2/2} = \mathbb{E}[e^{cY}]$, so

$$\mathbb{E}\big[e^{cY}\mathbf{1}_{Y>y_0}\big] = \mathbb{E}[e^{cY}]\;\Phi\!\left(\frac{\mu + c\tau^2 - y_0}{\tau}\right) = \mathbb{E}[s^{b/2}]\;\Phi\!\left(u + \frac{b\tau}{2}\right). \tag{12.6}$$

Dividing by $\mathbb{E}[s^{b/2}]$ and eliminating $u = \Phi^{-1}(q)$:

Lorenz curve of a lognormal envelope

$$L(q) = \Phi\!\left(\Phi^{-1}(q) + \frac{b\,\tau}{2}\right), \qquad \tau = \sqrt{\ln(\kappa/3)}, \qquad \alpha_{\max}(\rho) = \frac{1}{\Phi(\Phi^{-1}(\rho) - b\tau/2)}. \tag{12.7}$$

The expression of τ comes from (2.13): for a lognormal s, $\langle s^2\rangle / \langle s\rangle^2 = e^{\tau^2}$ (the same completion of the square with $c = 2$ and $c = 1$), so $\kappa = 3e^{\tau^2}$. The whole curve is set by the single number $b\tau/2$: the burstier the record (larger κ) or the larger the exponent, the higher the curve.

	$b = 5$		$b = 7$		kurtosis of the edited record	
kurtosis κ of the record	$\rho = 90\,\%$	$95\,\%$	$90\,\%$	$95\,\%$	$\alpha = 2$	$\alpha = 4$
4	2.1	1.6	3.6	2.5	3.5	3.3
6	4.7	3.0	19.5	9.8	4.5	4.0
9	11.1	6.1	118	47	6.1	5.0

Table 12.1: $\alpha_{\max}$ of (12.7) and the kurtosis of the edited record (Section 14) for a lognormal envelope. The large numbers at $b = 7$ are a warning as much as a result: the lognormal tail is heavy, and the closed form assumes one envelope common to all bands. On the example record, whose modulation is nearly common, the formula predicts 98, 92 and 82 % of the damage kept at $\alpha = 2$, 4 and 8, and the measured curves give 97, 88 and 79. On the thesis reference record, where the modulation depends on frequency (Section 12.5), the formula predicts 94 % at $\alpha = 4$ against a measured 77 % on average and much less at the worst oscillator. The closed form is an upper estimate; the measured curve of the worst oscillator is the limit.

12.5 What the kurtosis of the record says, and what it does not

Equation (2.13) gives the kurtosis of a modulated signal from its local variance, $\kappa = 3\,\langle s^2\rangle / \langle s\rangle^2$. Turned around, it is a test of the model: estimate $s(t)$ from the record (the running variance over short windows), form $3\,\langle s^2\rangle / \langle s\rangle^2$, and compare it with the kurtosis measured directly on the samples. If the two agree, the non-Gaussianity of the record is of the slow-modulation kind, one envelope common to all bands, and the closed form above applies. If the estimate from the local variance is clearly larger than the direct kurtosis, the bands are not modulated together: a

strong stationary band dilutes the direct kurtosis while the shocks that modulate the other bands dominate the running variance. This is the situation of Section 7, and it has two consequences for compression: a common envelope (kFDS, or the closed form) cannot describe the record, and the compression limit is stricter, because the stationary bands have a Lorenz curve close to the diagonal.

The running variance is itself a random estimate (Appendix C, C.3), so even a stationary Gaussian record gives an estimate above 3. The program removes this bias by computing the same estimate on a phase-randomised copy of the record (same PSD, no modulation) and subtracting its excess over 3. On the example record the corrected estimate is 6.10 against a direct kurtosis of 6.04; on the thesis reference record 7.8 against 6.5, and on seven channels of an industrial measurement (not distributed) the estimate exceeds the direct kurtosis by 0.4 to 1.9 in every channel. Real records are rarely modulated by one envelope. Their compression limits at 90 % retention were between 1.3 and 2.1 at the worst oscillator, against 3.0 for the example record.

13 The edit method

The method `edit` builds the test from the record itself: it keeps the stretches that carry the damage, drops the rest, and raises the amplitude only where, and only as much as, the editing could not keep the damage. It is the practical form of Section 12.2. Unlike the methods of Parts II and III it synthesises nothing: the kept stretches are samples of the record, with their waveform, their spectrum and their sequence of peaks.

13.1 Windows and their damage shares

The record is cut into M consecutive windows of length L_w. The default length is the larger of 2 s and three amplitude decay times of the slowest oscillator of the grid. The amplitude of a free response decays as $e^{-\zeta\omega_\mathrm{n}t}$, so its $1/e$ time is $1/(\zeta\omega_\mathrm{n}) = Q/(\pi f_\mathrm{n})$, twice the response time τ of (3.13) (which describes the variance). For $Q = 10$ and $f_\mathrm{min} = 5\,\mathrm{Hz}$ this gives $3 \times 0.64 = 1.9\,\mathrm{s}$, hence 2 s; for $Q = 20$ it gives 3.8 s. A window must be long enough for the response that a burst excites to lie mostly inside it, otherwise the damage of the burst is attributed to the next window; it must be short enough to separate the bursts from the quiet time, otherwise the Lorenz curve flattens. Within these bounds the result depends little on L_w.

For every oscillator i the response to the (pre-filtered, Section 10) record is computed once, cut into the windows, and the rainflow damage of each window is divided by the damage of the whole record. This gives the matrix of shares W_{ij} of Section 12.3. The damage of a window computed alone differs slightly from its contribution to the rainflow count of the whole response, because cycles that span a window boundary are closed differently; the shares are used to rank and to estimate, the final FDS is always measured on the assembled signal.

13.2 Ranking the windows

Two rankings are available.

Common. Windows are sorted by the sum of their shares over the oscillators, $\sum_i W_{ij}$. This keeps the largest damage on average and is the default.

Fair. Windows are added one at a time; each step adds the window that most raises the retention of the currently worst-served oscillator, $\min_i \sum_{j\in\mathrm{kept}} W_{ij}$ (greedy max–min). This equalises the retention over the oscillators when different bands want different windows (shocks at low f_n,

stationary content at high f_n), at the price of a lower average. On the records tested the fair ranking raised the Lorenz curve of the worst oscillator visibly but did not raise the damage of the assembled test, because where the damage is spread uniformly in time no ranking can help; the common ranking remains the default and the fair one an option.

13.3 Assembly: ring-down margin, cross-fades, time budget

The windows are taken in ranking order until the time budget T/α is used. Every kept window is extended by a ring-down margin of two amplitude decay times $2Q/(\pi f_{\min})$ of the slowest oscillator, so that the free response excited by a burst at the end of a window stays with the burst instead of being cut (a window that ends on a shock would otherwise keep the shock and lose the large cycles of its ring-down). Overlapping stretches are merged; when the next stretch would exceed the budget the selection stops, and single windows that still fit are added. The stretches are then joined in their original time order with raised-cosine cross-fades of 0.1 s (the fades overlap, and the budget accounts for the overlap). Keeping the original order keeps the slow trends of the record; a test of duration T/α therefore looks like the record with its quiet parts removed, which is what it is.

The realised compression differs slightly from the requested one (the budget is used in whole windows); it is reported. If the requested duration is not shorter than the record, the record is returned unchanged (with a warning when it is longer).

13.4 Life and repetitions

As for the other methods, a life T_L longer than the record is covered by playing the test r times (Section 4.3, with r the rounded ratio T_L/T of the target). The spectral gain of the next subsection then aims at the target FDS divided by r: the editing compresses the record, the repetitions cover the life, and the amplitude only makes up for the damage that the editing dropped.

13.5 Beyond the limit: the spectral gain

If the requested compression is below the limit (12.4), the assembled signal already carries the target damage at every oscillator to within a few percent and the method stops. Beyond the limit it is missing damage at some oscillators, typically those whose damage was spread uniformly in time. Raising the whole signal by one factor would also raise the oscillators that lost nothing. The program therefore applies a frequency-dependent gain $g(f)$, interpolated in $\log f$ between the nodes, as a zero-phase filter (multiplication of the Fourier transform, so no phase shift and no change of the peak sequence).

The damage at a node scales as the b-th power of the gain in its band (Section 5.4: $D \propto \sigma_z^b$ and $\sigma_z \propto g$). Writing $r_i = D_i/D_i^{\text{target}}$ for the measured ratio after a step, the gain of node i is updated as

$$\ln g_i \leftarrow \ln g_i - \frac{\omega_i}{b} \ln r_i, \tag{13.1}$$

with a relaxation ω_i that starts at 0.8 and is halved for a node whose error changes sign between two steps (so that two coupled nodes settle instead of oscillating), a cap of ± 0.15 on the step in $\ln g$, and a clip of g to $[1/g_{\max}, g_{\max}]$ (default $g_{\max} = 2$). Each step measures the exact rainflow FDS of the gained signal; the loop stops when every node is within the tolerance (default 5 %) or after the step budget. This is the closed loop of Section 5.4 with a gain instead of a PSD.

Two oscillators that are driven by the same input band cannot be set independently by any smooth gain. The usual case is the top of the grid above the last strong peak of the record: the responses of the nodes just above the peak are fed by the peak, not by their own band, and a gain between them raises both. The loop then ends in a compromise, the result table shows the two nodes off the target, and the program warns when the gain reaches its cap. The remedy is to end the grid where the record has content, or to accept the compromise.

What the gain does to the exponent dependence

After the gain the test is exact at the design exponent b at every node. At another exponent b' the damage of a node is off by the factor $g_i^{\,b'-b}$ times the (small) factor the editing itself produces. Where the editing kept the damage, $g_i \approx 1$ and the test is right for every exponent; where the amplitude did the work, the dependence on the exponent is exactly that of an amplitude scaling. In a plot of $\ln(D_{\text{test}}/D_{\text{record}})$ against b the curves are straight lines, and their slope $\ln g_{\text{eff}}$ measures how much a test relies on amplitude (Figure 2).

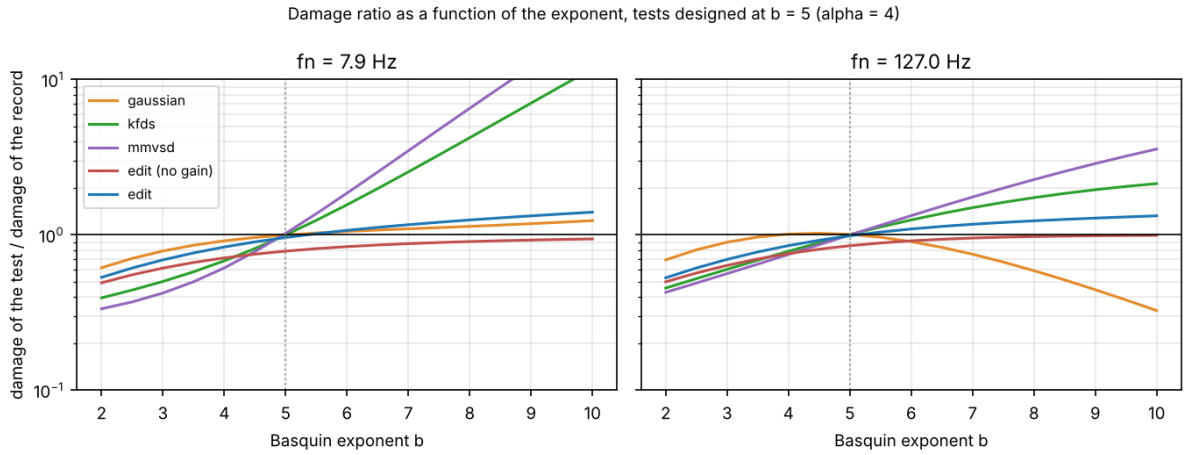


Figure 2: Example record compressed four times, tests designed at $b = 5$, $Q = 10$: damage of the test over damage of the record as a function of the exponent at two oscillators. Every synthesis method crosses 1 at the design exponent and departs from it as a power of its effective amplitude factor; the edited record without gain stays near 1 for $b \geq 5$ (and below 1 at small b, the small cycles it dropped); with the gain it is exact at $b = 5$ and inherits a mild slope where the gain acted.

13.6 Results

Table 13.1 gives the benchmark of `tools/compression_benchmark.py` on the example record and on the thesis reference record (the profile of Fig. 3.11 of the dissertation, used in the benchmarks of the README; the record itself is not distributed): compression $\alpha = 4$, 17 oscillators from 5 to 200 Hz at 1/3 octave, $Q = 10$, design $b = 5$, seed 1. The entries are the geometric mean over the oscillators of the ratio "damage of the test / damage of the record" at each exponent, and in brackets the ratio of the largest to the smallest value over the oscillators (1 would mean the same ratio at every node).

Record	Method	$b = 3$	$b = 5$ (design)	$b = 7$	$b = 9$
example, $\kappa = 6.0$, limit $\alpha_{\max} = 3.0$	`gaussian`	0.92 (1.4)	1.00 (1.1)	0.74 (1.9)	0.48 (4.0)
	`kfds`	0.56 (1.4)	1.00 (1.1)	1.77 (2.0)	2.96 (4.8)
	`mmvsd`	0.59 (1.7)	0.99 (1.1)	1.58 (3.0)	2.40 (9.0)
	`edit`, no gain	0.65 (1.2)	0.87 (1.2)	0.96 (1.1)	0.98 (1.1)
	`edit`	0.71 (1.1)	0.99 (1.0)	1.15 (1.1)	1.25 (1.2)
thesis record, $\kappa = 6.5$, limit $\alpha_{\max} = 1.6$	`gaussian`	1.49 (3.6)	1.00 (4.1)	0.46 (8.5)	0.19 (29)
	`kfds`	0.98 (3.7)	1.01 (4.3)	0.87 (11)	0.74 (82)
	`mmvsd`	0.63 (1.8)	0.95 (1.4)	1.18 (3.5)	1.34 (13)
	`edit`, no gain	0.52 (2.7)	0.72 (3.0)	0.82 (2.9)	0.86 (2.7)
	`edit`	0.64 (1.5)	1.00 (1.1)	1.29 (1.7)	1.55 (2.8)

Table 13.1: damage ratio test/record at four exponents for tests designed at $b = 5$, $\alpha = 4$.

Reading the table. At the design exponent every FDS method is right, by construction; the other columns separate them. On the example record, where the compression is only a little beyond the limit, the edited record keeps 87 % of the damage at $b = 5$ and 98 % at $b = 9$ with the same ratio at every oscillator; the gain adds the missing 13 % and a slope of 1.25 at $b = 9$. The synthesis methods give half of the damage ($\kappa_z \approx 3$ everywhere, the large cycles are missing) or two to three times the damage at $b = 9$ (the kurtosis they prescribe forces a heavier tail than the record has, Section 14). On the thesis record, whose damage below 100 Hz is carried by a few shocks while a stationary band around 160 Hz dominates the energy, the compression $\alpha = 4$ is well beyond the limit 1.6: the edited record keeps 72 % at $b = 5$, the gain works with factors up to 1.3 in the stationary band, and the $b = 9$ ratio is 1.55 with a spread of 2.8; the synthesis methods have spreads of 13 to 82, that is, oscillators where the test is off by one or two orders of magnitude when the exponent changes, mostly at the top of the grid where the record has little content of its own. The edited record without gain is the only profile whose ratio has the same order at every oscillator and every exponent; it is also the only one that is short of the target rather than beyond it.

Two further observations from the same study. First, the response kurtosis κ_z of the edited test is lower than the record's at the oscillators where quiet time was removed (Section 14); MMVSD, which prescribes κ_z, matches it best, as designed. Second, on a nonlinear oscillator (a hardening spring, cubic term equal to half the linear one at three times the RMS response) the edited record transferred the damage with a fidelity of 0.75 to 1.0 relative to its linear fidelity on both records, while the synthesis methods ranged from 0.35 to 2.7, in both directions depending on the record: a synthesised signal matches the linear statistics it was designed for, an edited record carries the waveform. This is one toy nonlinearity and should be read as an indication.

14 Kurtosis under compression

14.1 The kurtosis of an edited record

The kurtosis of a signal compares its fourth moment with the square of its variance. Removing the quiet stretches of a record raises the variance of what remains; whether it raises the fourth moment as much depends on where the fourth moment sits. For a modulated signal the computation is immediate with the local variance of Section 2.5: keeping the stretches with s above a

threshold, the moments of what is kept are the conditional moments of s, and by (2.13)

$$\kappa_{\text{edited}} = 3\,\frac{\langle s^2 \mid \text{kept}\rangle}{\langle s \mid \text{kept}\rangle^2}. \tag{14.1}$$

For the lognormal envelope of Section 12.4 the conditional moments are the truncated expectations computed there, with $c = 2$ and $c = 1$:

$$\kappa_{\text{edited}}(q) = 3\,e^{\tau^2}\,\frac{\Phi(u+2\tau)\,\Phi(u)}{\Phi(u+\tau)^2}, \qquad u = \Phi^{-1}(q), \tag{14.2}$$

which tends to $3e^{\tau^2} = \kappa$ when everything is kept ($q \to 1$) and decreases as the test is shortened (last two columns of Table 12.1): a record with kurtosis 6 edited to a quarter of its length has kurtosis 4.0. The measured values agree: 6.04 to 4.5 on the example record. The reason is the one stated above: the quiet time contributed little to the fourth moment but much to the normalisation.

The decrease is a property of smooth modulation. When the kurtosis of a record comes from a few very large shocks, the editing keeps the shocks, the removed quiet time raises the variance only moderately, and the kurtosis hardly changes: 6.47 to 6.5 on the thesis reference record. The two records that look alike by their kurtosis behave differently under compression, which is one more reason to look at the Lorenz curve rather than at the kurtosis.

14.2 The kurtosis target of a compressed test

The methods that prescribe the kurtosis (kFDS, RF, and the kurtosis control methods) use by default the kurtosis of the target record. Section 13.6 showed what this does under compression: a test that must carry the damage of the record in a quarter of the time *and* keep the record's kurtosis is forced to a heavier tail than the record has, exact at the design exponent and two to three times too damaging at $b = 9$ on the example record. The edited record shows the natural target: the kurtosis that the damaging part of the record has once the quiet time is gone.

Compressed kurtosis target

For a test of duration T/α built from a record, the kurtosis to prescribe is the kurtosis of the record edited to T/α (Section 13, without gain), not the kurtosis of the whole record. The option `compress_kurtosis` (interface: "compressed kurtosis target"; command line: `--compress-kurtosis`) computes it and uses it as the kurtosis target of kFDS, RF, MLV, VSD and PM.

On the example record this single change brought the $b = 9$ ratio of kFDS from 2.96 to 1.7 and halved its distance from the record's cycle histogram in the tail; on the thesis record, whose kurtosis does not drop under editing, it changed nothing, and on seven other channels it improved kFDS in most cases and worsened it in two where the record's kurtosis was low to begin with. It is an option, not a default: when the Lorenz curve says that a record is compressible, it is the right target; when the kurtosis comes from shocks it is neutral.

The same reasoning applies to the response kurtosis spectrum $\kappa_z(f_\text{n})$ that MMVSD prescribes. The κ_z of the edited record is lower than that of the whole record at the oscillators where quiet time was removed, and it is the κ_z that a compressed test should carry. Version 2.2 does not yet pass it to MMVSD automatically; it can be done by hand by analysing an `edit` result (no gain) as a record and using it as the MMVSD target.

15 The response kurtosis spectrum and the non-stationarity matrix

Part III prescribes the kurtosis of the response of every oscillator, $\kappa_z(f_n)$, because it is the quantity that governs how the damage is produced (Section 4.6). This section places it in a slightly larger frame, the two-frequency description of non-stationarity introduced by Trapp and Wolfsteiner (Appendix C, C.4), shows what the frame adds and what it costs, and how it relates to the MMVSD model.

15.1 Fourth moment of a response from the local spectrum

Let the local PSD of the record be $G(f,t) = P(f)\, m(f,t)$ with $\langle m(f,\cdot)\rangle = 1$, the evolutionary spectrum of Section 7.2. Assume, as in Section 7, that the modulation is slow compared with the response times, so that at every instant the response of an oscillator is Gaussian with the local variance

$$s(t) = \sum_j W_j\, G(f_j,t)\, \Delta f_j \tag{15.1}$$

(Lalanne's operator (5.1) applied to the local spectrum, written on a frequency grid f_j with bin widths Δf_j; $W_j = |H(f_j)|^2$ is the row of the oscillator). A locally Gaussian response has the fourth moment $3s(t)^2$ (Section 2.2), so its time average is

$$\left\langle z^4 \right\rangle = 3\, \left\langle s^2 \right\rangle = 3 \sum_{j,k} W_j W_k\, \Delta f_j \Delta f_k\ \left\langle G(f_j,t)\, G(f_k,t) \right\rangle . \tag{15.2}$$

The double sum contains the time average of a product of two local spectral values. Dividing it by the product of the mean spectra isolates the non-stationarity:

Non-stationarity matrix

$$N_{jk} = \frac{\langle G(f_j,t)\, G(f_k,t)\rangle}{P(f_j)\, P(f_k)} - 1 = \langle m(f_j,t)\, m(f_k,t)\rangle - 1, \tag{15.3}$$

the covariance of the modulation between two bands. It is zero for a stationary signal. With the weights $w_j = W_j P(f_j)\Delta f_j / \langle s\rangle$ (the share of band j in the variance of the response, $\sum_j w_j = 1$), the kurtosis of the response is

$$\kappa_z = \frac{\left\langle z^4 \right\rangle}{\langle s\rangle^2} = 3 + 3 \sum_{j,k} w_j\, N_{jk}\, w_k = 3 + 3\, w^{\mathsf{T}} N w. \tag{15.4}$$

The derivation is the one of Section 7.4 written for an arbitrary filter instead of the oscillator family: the matrix is the second-order statistics of the MMVSD modulation field $m(f,t)$ itself. Its diagonal $N_{jj} = \mathrm{Var}[m(f_j,\cdot)]$ is the depth of the modulation of one band, the quantity that MMVSD calls the excess; its off-diagonal elements say whether two bands pulsate together (positive), independently (zero) or in opposition (negative), which in the MMVSD field is set by the burst width along frequency (Section 7.7).

Two limiting cases make the matrix concrete. One envelope for all bands, $m(f,t) = e(t)^2$, gives $N_{jk} = \left\langle e^4 \right\rangle / \left\langle e^2 \right\rangle^2 - 1 = \nu$ for every pair: a matrix of rank one, and $\kappa_z = 3 + 3\nu \sum_{jk} w_j w_k = 3 + 3\nu$ for every filter whatsoever, the result of Section 2.5. Independent bands give a diagonal matrix and $\kappa_z = 3 + 3\sum_j w_j^2 N_{jj}$: a narrow filter that sees one band receives the full depth of that band, a wide filter or a structure with several modes averages over many bands and its $\sum_j w_j^2$ is small, so its response is nearly Gaussian even when every single band pulsates strongly. Independent modulation averages out, common modulation does not.

15.2 What the matrix adds to $\kappa_z(f_\mathrm{n})$

Equation (15.4) says that the curve $\kappa_z(f_\mathrm{n})$ is a one-dimensional projection of a two-dimensional object: one quadratic form of the matrix for each member of one family of filters (the oscillators of the grid with the chosen Q). The matrix predicts κ_z for any linear filter: a different Q, a structure with several modes, two coupled modes whose cross terms N_{jk} do not appear in the curve at all. Conversely, the matrix cannot be recovered from the curve: many matrices give the same $\kappa_z(f_\mathrm{n})$, and two records with the same curve can load a multi-mode component differently. For the oscillator family that the FDS uses, the curve is the quantity tied to the damage and the one MMVSD prescribes; the matrix is the diagnostic that says whether the bands of a record move together (rank one: kFDS is enough) or separately (MMVSD is needed), and the right target when the component is not a single oscillator.

15.3 Estimation

The matrix is estimated from a spectrogram (Section 11): the local spectra of the segments are the $G(f_j, t)$, and (15.3) is a sample covariance over the segments. Two cautions. It is a fourth-order statistic, so it needs long records to be stable. And the spectrogram is itself random: for a stationary Gaussian record a periodogram value fluctuates with a relative variance of 1 (Appendix C, C.3), so the raw estimate is positive on the diagonal and near it even without any modulation. The program's diagnostic removes this bias by computing the same estimate on a phase-randomised copy of the record (same PSD, random Fourier phases, hence stationary and Gaussian) and subtracting it. On the example record the corrected matrix shows a common background near 0.8 (the road level changes, modulating all bands together), stronger diagonal blocks at the body resonances (the potholes excite them separately), and the $\kappa_z(f_\mathrm{n})$ predicted from it through (15.4) follows the measured curve including its two peaks, which the rank-one (common envelope) assumption cannot produce. The prediction overshoots at low natural frequencies and at the 12 Hz peak: there the response time of the oscillator is comparable with the duration of the shocks, part of the modulation is averaged out (Section 3.6), and the measured κ_z stays below the value a slowly varying spectrum would give. The matrix gives an upper bound in that regime, the measured curve the truth.

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A Lalanne's operator in closed form

This appendix integrates (3.10): the response variance of an oscillator to a PSD that is linear between nodes.

A.1 Dimensionless form

Fix the oscillator of natural frequency f_{n} and write every frequency relative to it, $h = f/f_{\mathrm{n}}$. Then $\omega = h\omega_{\mathrm{n}}$, and (3.5) becomes

$$|H|^2 = \frac{1}{(\omega_{\mathrm{n}}^2 - h^2\omega_{\mathrm{n}}^2)^2 + (2\zeta\omega_{\mathrm{n}} \cdot h\omega_{\mathrm{n}})^2} = \frac{1}{\omega_{\mathrm{n}}^4\,\Delta(h)}, \qquad \Delta(h) = (1-h^2)^2 + (2\zeta h)^2. \tag{A.1}$$

With $\mathrm{d}f = f_{\mathrm{n}}\,\mathrm{d}h$, the variance is

$$\sigma_z^2 = \int |H|^2 G\,\mathrm{d}f = \frac{f_{\mathrm{n}}}{\omega_{\mathrm{n}}^4}\int \frac{G(f_{\mathrm{n}}h)}{\Delta(h)}\,\mathrm{d}h. \tag{A.2}$$

A.2 Factorisation of Δ

Expanding, $\Delta(h) = 1 - 2h^2 + h^4 + 4\zeta^2h^2 = (h^2+1)^2 - (4-4\zeta^2)h^2$. With

$$a = 2\sqrt{1-\zeta^2} \qquad (a^2 = 4 - 4\zeta^2), \tag{A.3}$$

this is a difference of squares, $\Delta = (h^2+1)^2 - (ah)^2$:

$$\Delta(h) = (h^2 + ah + 1)(h^2 - ah + 1). \tag{A.4}$$

Both factors are positive: $h^2 \pm ah + 1 = (h \pm a/2)^2 + 1 - a^2/4$ and $1 - a^2/4 = 1 - (1-\zeta^2) = \zeta^2$, so

$$h^2 \pm ah + 1 = \left(h \pm \frac{a}{2}\right)^2 + \zeta^2. \tag{A.5}$$

A.3 Partial fractions

Since $h \to -h$ exchanges the two factors, look for

$$\frac{1}{\Delta} = \frac{\alpha h + \beta}{h^2 + ah + 1} + \frac{-\alpha h + \beta}{h^2 - ah + 1}. \tag{A.6}$$

Over the common denominator the numerator is $(\alpha h+\beta)(h^2-ah+1) + (-\alpha h+\beta)(h^2+ah+1)$. Expanding both products, $\alpha h^3 - a\alpha h^2 + \alpha h + \beta h^2 - a\beta h + \beta$ and $-\alpha h^3 - a\alpha h^2 - \alpha h + \beta h^2 + a\beta h + \beta$; their sum is $(2\beta - 2a\alpha)h^2 + 2\beta$. It must equal 1 for all h: $2\beta = 1$ and $2\beta = 2a\alpha$, so $\beta = \frac{1}{2}$ and $\alpha = \frac{1}{2a}$:

$$\frac{1}{\Delta} = \frac{\frac{h}{2a} + \frac{1}{2}}{h^2 + ah + 1} + \frac{-\frac{h}{2a} + \frac{1}{2}}{h^2 - ah + 1}. \tag{A.7}$$

A.4 The two basic integrals

First integral. In the first fraction write the numerator as a multiple of the derivative of the denominator plus a constant: $\frac{h}{2a} + \frac{1}{2} = \frac{2h+a}{4a} + \frac{1}{4}$ (indeed $\frac{2h}{4a} + \frac{a}{4a} + \frac{1}{4} = \frac{h}{2a} + \frac{1}{2}$). The derivative of $h^2 + ah + 1$ is $2h + a$, and by (A.5)

$$\int \frac{\mathrm{d}h}{h^2 + ah + 1} = \int \frac{\mathrm{d}h}{(h + \frac{a}{2})^2 + \zeta^2} = \frac{1}{\zeta}\arctan\frac{h + \frac{a}{2}}{\zeta} = \frac{1}{\zeta}\arctan\frac{2h+a}{2\zeta}. \tag{A.8}$$

Hence $\int \frac{\frac{h}{2a}+\frac{1}{2}}{h^2+ah+1}\mathrm{d}h = \frac{1}{4a}\ln(h^2+ah+1) + \frac{1}{4\zeta}\arctan\frac{2h+a}{2\zeta}$. The second fraction is treated in the same way with $-\frac{h}{2a} + \frac{1}{2} = -\frac{2h-a}{4a} + \frac{1}{4}$, giving $-\frac{1}{4a}\ln(h^2 - ah + 1) + \frac{1}{4\zeta}\arctan\frac{2h-a}{2\zeta}$. Adding,

$$\int \frac{\mathrm{d}h}{\Delta} = \frac{1}{4a}\ln\frac{h^2+ah+1}{h^2-ah+1} + \frac{1}{4\zeta}\left[\arctan\frac{2h+a}{2\zeta} + \arctan\frac{2h-a}{2\zeta}\right]. \tag{A.9}$$

Second integral. Substitute $v = h^2$, $\mathrm{d}v = 2h\,\mathrm{d}h$:

$$\int \frac{h\,\mathrm{d}h}{\Delta} = \frac{1}{2}\int \frac{\mathrm{d}v}{(1-v)^2 + 4\zeta^2 v} = \frac{1}{2}\int \frac{\mathrm{d}v}{v^2 + (4\zeta^2-2)v + 1}. \tag{A.10}$$

Complete the square: $(v + 2\zeta^2 - 1)^2 = v^2 + (4\zeta^2-2)v + (2\zeta^2-1)^2$, and $1 - (2\zeta^2-1)^2 = 1 - 4\zeta^4 + 4\zeta^2 - 1 = 4\zeta^2(1-\zeta^2) = \zeta^2 a^2$, so the denominator is $(v + 2\zeta^2 - 1)^2 + \zeta^2 a^2$ and

$$\int \frac{h\,\mathrm{d}h}{\Delta} = \frac{1}{2\zeta a}\arctan\frac{h^2 + 2\zeta^2 - 1}{\zeta a}. \tag{A.11}$$

Both antiderivatives are continuous in h. It is convenient to normalise them by $4\zeta/\pi$:

$$I_0(h) = \frac{4\zeta}{\pi}\int \frac{\mathrm{d}h}{\Delta} = \frac{\zeta}{\pi a}\ln\frac{h^2 + ah + 1}{h^2 - ah + 1} + \frac{1}{\pi}\left[\arctan\frac{2h + a}{2\zeta} + \arctan\frac{2h - a}{2\zeta}\right], \tag{A.12}$$

$$I_1(h) = \frac{4\zeta}{\pi}\int \frac{h\,\mathrm{d}h}{\Delta} = \frac{2}{\pi a}\arctan\frac{h^2 + 2\zeta^2 - 1}{\zeta a}. \tag{A.13}$$

(The program writes I_1 as $\frac{2}{\pi a}[\arctan\frac{2h-a}{2\zeta} - \arctan\frac{2h+a}{2\zeta}]$, which has the same derivative and therefore differs only by a constant, irrelevant in definite integrals.)

White noise. At $h = 0$: $\ln 1 = 0$ and $\arctan\frac{a}{2\zeta} + \arctan\frac{-a}{2\zeta} = 0$, so $I_0(0) = 0$. For $h \to \infty$ the logarithm tends to $\ln 1 = 0$ and each arctangent to $\pi/2$, so $I_0(\infty) = 1$. Therefore $\int_0^\infty \mathrm{d}h/\Delta = \pi/(4\zeta)$, and by (A.2) with $G = G_0$

$$\int_0^\infty |H|^2 \mathrm{d}f = \frac{f_\mathrm{n}}{\omega_\mathrm{n}^4}\,\frac{\pi}{4\zeta} = \frac{\omega_\mathrm{n}}{2\pi}\cdot\frac{\pi}{4\zeta\omega_\mathrm{n}^4} = \frac{1}{8\zeta\omega_\mathrm{n}^3}, \tag{A.14}$$

the result used in (3.9).

A.5 A PSD linear between nodes

Let the nodes, in units of the oscillator's natural frequency, be $h_l = f_{\mathrm{n},l}/f_\mathrm{n}$. On the segment $[h_l, h_{l+1}]$ of width $\delta_l = h_{l+1} - h_l$ the PSD is

$$G = G_l\,\frac{h_{l+1} - h}{\delta_l} + G_{l+1}\,\frac{h - h_l}{\delta_l}. \tag{A.15}$$

With the increments $\Delta I_{0,l} = I_0(h_{l+1}) - I_0(h_l)$ and $\Delta I_{1,l} = I_1(h_{l+1}) - I_1(h_l)$,

$$\frac{4\zeta}{\pi}\int_{h_l}^{h_{l+1}} \frac{G}{\Delta}\mathrm{d}h = G_l\,\frac{h_{l+1}\Delta I_{0,l} - \Delta I_{1,l}}{\delta_l} + G_{l+1}\,\frac{\Delta I_{1,l} - h_l \Delta I_{0,l}}{\delta_l}. \tag{A.16}$$

The node j appears as the right end of segment $j-1$ and as the left end of segment j. Collecting its two coefficients and multiplying by $\frac{f_\mathrm{n}}{\omega_\mathrm{n}^4}\frac{\pi}{4\zeta} = \frac{1}{8\zeta\omega_\mathrm{n}^3}$ (from (A.2)),

Lalanne's operator

$$W_{ij} = \frac{1}{8\zeta\omega_\mathrm{n}^3}\left[\frac{\Delta I_{1,j-1} - h_{j-1}\Delta I_{0,j-1}}{\delta_{j-1}} + \frac{h_{j+1}\Delta I_{0,j} - \Delta I_{1,j}}{\delta_j}\right], \tag{A.17}$$

where ω_n, h and the increments refer to the oscillator i, and the first (second) term is absent for the first (last) node. The program evaluates this for all pairs (i, j) at once. The moment operators of MMVSD (Section 7.3) include the shape prior, the digital response and the edge extension, which have no closed form; they are integrated numerically, and with $\Pi = 1$ and the analogue response they reproduce this matrix to better than 10^{-3} (checked in the tests).

B The ramp-invariant oscillator filter

This appendix derives the coefficients of the recursion (3.16) (the ramp-invariant filter, after Smallwood). Throughout, $\omega = \omega_{\mathrm{n}}$, $\omega_{\mathrm{d}} = \omega\sqrt{1-\zeta^2}$ is the damped natural angular frequency, and the symbol q denotes the variable of the z-transform (the letter z is the relative displacement).

B.1 The input between samples

Assume that the base acceleration varies linearly between consecutive samples. With the triangle function $\Lambda(\theta) = \max(0, 1-|\theta|)$, which is 1 at $\theta = 0$, 0 at $\theta = \pm 1$ and linear in between,

$$x(t) = \sum_k x_k \, \Lambda\!\left(\frac{t}{\Delta t} - k\right) \tag{B.1}$$

is exactly the linear interpolation of the samples. The oscillator is linear and time invariant, so its response at the sampling instants is

$$z_l = z(l\Delta t) = \sum_k x_k \, \gamma_{l-k}, \tag{B.2}$$

where γ_l is the response, at $t = l\Delta t$, to the single triangle $\Lambda(t/\Delta t)$ centred at $t = 0$, starting from rest.

B.2 The triangle as a combination of ramps

Let $r(t) = t$ for $t \geq 0$ and 0 for $t < 0$ (a unit ramp starting at 0). Then

$$\Lambda(t/\Delta t) = \frac{1}{\Delta t}\big[r(t+\Delta t) - 2r(t) + r(t-\Delta t)\big]. \tag{B.3}$$

Check on the four intervals: for $t \leq -\Delta t$ all ramps are zero; for $-\Delta t \leq t \leq 0$ only the first is active and gives $(t+\Delta t)/\Delta t$, rising from 0 to 1; for $0 \leq t \leq \Delta t$ it gives $(t+\Delta t-2t)/\Delta t = (\Delta t - t)/\Delta t$, falling from 1 to 0; for $t \geq \Delta t$ it gives $(t+\Delta t-2t+t-\Delta t)/\Delta t = 0$. By linearity, if $\rho(t)$ is the response to the ramp $r(t)$,

$$\gamma_l = \frac{1}{\Delta t}\big[\rho_{l+1} - 2\rho_l + \rho_{l-1}\big], \qquad \rho_l = \rho(l\Delta t), \tag{B.4}$$

with $\rho_l = 0$ for $l \leq 0$.

B.3 The ramp response

ρ solves $\ddot{\rho} + 2\zeta\omega\dot{\rho} + \omega^2\rho = t$ for $t \geq 0$ with $\rho(0) = \dot{\rho}(0) = 0$. A particular solution is $\rho_p = t/\omega^2 - 2\zeta/\omega^3$: indeed $\dot{\rho}_p = 1/\omega^2$, $\ddot{\rho}_p = 0$, and $2\zeta\omega/\omega^2 + \omega^2(t/\omega^2 - 2\zeta/\omega^3) = 2\zeta/\omega + t - 2\zeta/\omega = t$. The homogeneous solutions are $e^{-\zeta\omega t}(A\cos\omega_{\mathrm{d}}t + B\sin\omega_{\mathrm{d}}t)$. The initial conditions give

$$\rho(0) = -\frac{2\zeta}{\omega^3} + A = 0 \qquad \Rightarrow \quad A = \frac{2\zeta}{\omega^3}, \tag{B.5}$$

$$\dot{\rho}(0) = \frac{1}{\omega^2} - \zeta\omega A + \omega_{\mathrm{d}}B = 0 \qquad \Rightarrow \quad B = \frac{\zeta\omega A - 1/\omega^2}{\omega_{\mathrm{d}}} = \frac{2\zeta^2 - 1}{\omega^2\omega_{\mathrm{d}}}. \tag{B.6}$$

Hence, for $t \geq 0$,

$$\rho(t) = \frac{t}{\omega^2} - \frac{2\zeta}{\omega^3} + e^{-\zeta\omega t}\left[\frac{2\zeta}{\omega^3}\cos\omega_{\mathrm{d}}t + \frac{2\zeta^2-1}{\omega^2\omega_{\mathrm{d}}}\sin\omega_{\mathrm{d}}t\right]. \tag{B.7}$$

B.4 z-transforms

Write $E = e^{-\zeta\omega\Delta t}$, $c = \cos\omega_{\mathrm{d}}\Delta t$, $s = \sin\omega_{\mathrm{d}}\Delta t$ and $w = q^{-1}$. The transform of a sequence u_l ($l \geq 0$) is $U(q) = \sum_{l\geq 0} u_l q^{-l}$. Four transforms are needed.

- $u_l = 1$: the geometric series $\sum_l w^l = \frac{1}{1-w}$.
- $u_l = l\Delta t$: differentiating the geometric series, $\sum_l l w^l = w\frac{\mathrm{d}}{\mathrm{d}w}\frac{1}{1-w} = \frac{w}{(1-w)^2}$, so the transform is $\frac{\Delta t\, w}{(1-w)^2}$.
- $u_l = E^l\cos(l\omega_{\mathrm{d}}\Delta t)$ and $u_l = E^l\sin(l\omega_{\mathrm{d}}\Delta t)$: they are the real and imaginary parts of $(Ee^{\mathrm{j}\omega_{\mathrm{d}}\Delta t})^l$, whose transform is $\frac{1}{1-Ee^{\mathrm{j}\omega_{\mathrm{d}}\Delta t}w}$. Multiplying numerator and denominator by $1 - Ee^{-\mathrm{j}\omega_{\mathrm{d}}\Delta t}w$, the denominator becomes $(1 - Ee^{\mathrm{j}\omega_{\mathrm{d}}\Delta t}w)(1 - Ee^{-\mathrm{j}\omega_{\mathrm{d}}\Delta t}w) = 1 - 2Ecw + E^2w^2 =: \mathcal{D}(w)$ and the numerator $1 - Ee^{-\mathrm{j}\omega_{\mathrm{d}}\Delta t}w = 1 - Ecw + \mathrm{j}Esw$. Therefore the transforms are $\frac{1-Ecw}{\mathcal{D}(w)}$ and $\frac{Esw}{\mathcal{D}(w)}$.

Applying them to ρ_l:

$$R(q) = \frac{\Delta t\, w}{\omega^2(1-w)^2} - \frac{2\zeta}{\omega^3}\frac{1}{1-w} + \frac{\mathcal{N}(w)}{\mathcal{D}(w)}, \qquad \mathcal{N}(w) = n_0 + n_1 w, \tag{B.8}$$

with

$$n_0 = \frac{2\zeta}{\omega^3}, \qquad n_1 = -\frac{2\zeta}{\omega^3}Ec + \frac{2\zeta^2-1}{\omega^2\omega_{\mathrm{d}}}Es. \tag{B.9}$$

B.5 The transfer function

By (B.4), the transform of γ_l is $\Gamma_d(q) = \frac{1}{\Delta t}(q - 2 + q^{-1})R(q)$ (a shift by one sample to the left multiplies a transform by q, to the right by q^{-1}). Since $q - 2 + q^{-1} = q(1-w)^2$:

$$\Delta t\,\Gamma_d(q) = q(1-w)^2R(q) = \frac{\Delta t}{\omega^2} - \frac{2\zeta}{\omega^3}(q-1) + q(1-w)^2\frac{\mathcal{N}(w)}{\mathcal{D}(w)}, \tag{B.10}$$

where $qw = 1$ was used in the first two terms ($q(1-w)^2\cdot\frac{w}{(1-w)^2} = 1$ and $q(1-w)^2\cdot\frac{1}{1-w} = q(1-w) = q-1$). Over the common denominator $\mathcal{D}$, the numerator is

$$\frac{\Delta t}{\omega^2}\mathcal{D} - \frac{2\zeta}{\omega^3}(q-1)\mathcal{D} + (q-2+w)(n_0+n_1w). \tag{B.11}$$

The two products containing q expand as follows (using $q\mathcal{D} = q - 2Ec + E^2w$):

$$(q-1)\mathcal{D} = q - (1+2Ec) + (E^2+2Ec)w - E^2w^2, \tag{B.12}$$

$$(q-2+w)(n_0+n_1w) = n_0q + (n_1-2n_0) + (n_0-2n_1)w + n_1w^2. \tag{B.13}$$

The terms in q are $-\frac{2\zeta}{\omega^3}q + n_0q = 0$: they cancel, and the filter is causal. Collecting the powers of w:

$$w^0: \quad \frac{\Delta t}{\omega^2} + \frac{2\zeta}{\omega^3}(1+2Ec) + n_1 - 2n_0 = \frac{\Delta t}{\omega^2} + \frac{2\zeta}{\omega^3}(Ec-1) + \frac{2\zeta^2-1}{\omega^2\omega_{\mathrm{d}}}Es, \tag{B.14}$$

$$w^1: \quad -\frac{2Ec\,\Delta t}{\omega^2} - \frac{2\zeta}{\omega^3}(E^2+2Ec) + n_0 - 2n_1 = -\frac{2Ec\,\Delta t}{\omega^2} + \frac{2\zeta}{\omega^3}(1-E^2) - 2\,\frac{2\zeta^2-1}{\omega^2\omega_{\mathrm{d}}}Es, \tag{B.15}$$

$$w^2: \quad \frac{E^2\Delta t}{\omega^2} + \frac{2\zeta}{\omega^3}E^2 + n_1 = \frac{E^2\Delta t}{\omega^2} + \frac{2\zeta}{\omega^3}(E^2-Ec) + \frac{2\zeta^2-1}{\omega^2\omega_{\mathrm{d}}}Es. \tag{B.16}$$

(In the w^0 line: $\frac{2\zeta}{\omega^3}(1+2Ec) - 2n_0 - \frac{2\zeta}{\omega^3}Ec = \frac{2\zeta}{\omega^3}(1+2Ec-2-Ec) = \frac{2\zeta}{\omega^3}(Ec-1)$. In the w^1 line the terms $\mp\frac{4\zeta}{\omega^3}Ec$ cancel.) Dividing by Δt and writing every coefficient over $\omega^3\Delta t$:

Ramp-invariant coefficients

$$b_1 = \frac{\omega\Delta t + 2\zeta(Ec-1) + \frac{\omega}{\omega_\mathrm{d}}(2\zeta^2-1)Es}{\omega^3\Delta t}, \tag{B.17}$$

$$b_2 = \frac{-2\omega\Delta t\,Ec + 2\zeta(1-E^2) - 2\frac{\omega}{\omega_\mathrm{d}}(2\zeta^2-1)Es}{\omega^3\Delta t}, \tag{B.18}$$

$$b_3 = \frac{(2\zeta+\omega\Delta t)E^2 + E\big(\frac{\omega}{\omega_\mathrm{d}}(2\zeta^2-1)s - 2\zeta c\big)}{\omega^3\Delta t}, \tag{B.19}$$

$$a_1 = -2Ec, \qquad a_2 = E^2, \tag{B.20}$$

and $\Gamma_d(q) = (b_1 + b_2q^{-1} + b_3q^{-2})/(1 + a_1q^{-1} + a_2q^{-2})$. By (B.2) the sequence z_l is the input sequence filtered by Γ_d, which is the recursion (3.16).

Check. At zero frequency ($q = 1$) the gain must be the static deflection $1/\omega^2$. Numerically, for $f_\mathrm{n} = 20\,\mathrm{Hz}$, $Q = 10$, $f_s = 500\,\mathrm{Hz}$, $(b_1 + b_2 + b_3)/(1 + a_1 + a_2) = 6.33257 \times 10^{-5}$ and $1/\omega^2 = 6.33257 \times 10^{-5}$. Inserting $q = e^{\mathrm{j}\theta}$ gives the frequency response (3.17); the program evaluates it in closed form for all nodes and frequencies at once and checks it against a direct evaluation of the polynomials.

C Results quoted from the literature

C.1 Dirlik's distribution of rainflow ranges

Dirlik fitted, on a large set of simulated Gaussian processes, a closed-form probability density of the rainflow ranges S of a stationary Gaussian process with the spectral moments m_0, m_1, m_2, m_4 (Section 3.3). With

$$x_m = \frac{m_1}{m_0}\sqrt{\frac{m_2}{m_4}}, \qquad \gamma = \frac{m_2}{\sqrt{m_0 m_4}}, \tag{C.1}$$

$$D_1 = \frac{2(x_m - \gamma^2)}{1+\gamma^2}, \qquad R = \frac{\gamma - x_m - D_1^2}{1-\gamma-D_1+D_1^2}, \qquad D_2 = \frac{1-\gamma-D_1+D_1^2}{1-R}, \tag{C.2}$$

$$D_3 = 1 - D_1 - D_2, \qquad Q_D = \frac{1.25(\gamma - D_3 - D_2R)}{D_1}, \tag{C.3}$$

the density of the normalised range $Z = S/(2\sqrt{m_0})$ is

$$p(Z) = \frac{D_1}{Q_D}e^{-Z/Q_D} + \frac{D_2\,Z}{R^2}e^{-Z^2/(2R^2)} + D_3\,Z\,e^{-Z^2/2}, \tag{C.4}$$

and the cycles occur at the rate of the peaks, $\nu_p = \sqrt{m_4/m_2}$. The damage uses the amplitude $A = S/2 = \sqrt{m_0}\,Z$. Its moment of order b follows term by term: for the exponential term the substitution $u = Z/Q_D$ gives $\int_0^\infty Z^b \frac{D_1}{Q_D}e^{-Z/Q_D}\mathrm{d}Z = D_1Q_D^b\Gamma(1+b)$; the other two terms are Rayleigh densities with $\sigma = R$ and $\sigma = 1$ (weighted by D_2 and D_3), whose moments are given by (4.8). Hence the expected damage per second

$$d_\mathrm{Dir} = \frac{K^b}{C}\,\nu_p\,m_0^{b/2}\Big[D_1Q_D^b\,\Gamma(1+b) + 2^{b/2}\,\Gamma\big(1+\tfrac{b}{2}\big)\,\big(D_2|R|^b + D_3\big)\Big]. \tag{C.5}$$

For a narrow-band process ($\gamma \to 1$) the formula tends to the Rayleigh result $\nu_0^+(2m_0)^{b/2}\Gamma(1+b/2)$ of (4.9) (checked numerically in the tests within 2 % for a pure tone spectrum). The program evaluates the formula with safeguards against divisions by zero for nearly pure tones.

C.2 Central limit theorem

A sum of many independent random variables with finite variances, none of which dominates the sum, is approximately Gaussian. Section 2.4 applies it to the lines of a random-phase multisine.

C.3 Welch's PSD estimate

Welch's estimate averages the periodograms of overlapping windowed segments. Its expectation is the true PSD smoothed over the resolution of a segment, and its relative scatter decreases as the square root of the number of independent segments. The program uses segments of length equal to the inverse of the requested resolution, a Hamming window and 50 % overlap.

C.4 The non-stationarity matrix

Trapp and Wolfsteiner (*International Journal of Fatigue* 146, 2021, 106115; see also their AP-SIPA 2021 paper on integrated spectral kurtosis) describe a varying random vibration load by a two-frequency decomposition of its fourth-order moment, the non-stationarity matrix, which tells whether frequency bands are modulated synchronously, and from which the kurtosis of the response of a linear system follows. Section 15 derives the same structure from the slowly varying spectrum of the MMVSD model; the exact normalisation and the estimators of the original are in the cited paper.

C.5 Non-stationary against non-Gaussian excitation

Česnik, Slavič, Capponi, Palmieri, Cianetti and Boltežar (MATEC Web of Conferences 165, 10011, 2018) measured times to failure under stationary non-Gaussian and under amplitude-modulated non-stationary excitations with the same PSD. The stationary non-Gaussian excitation did not change the time to failure appreciably; the modulated one shortened it markedly, and the amount depended on the modulation rate. This is the experimental counterpart of Sections 3.6 and 4.6: only the modulation that the oscillator can follow reaches the response and the damage.